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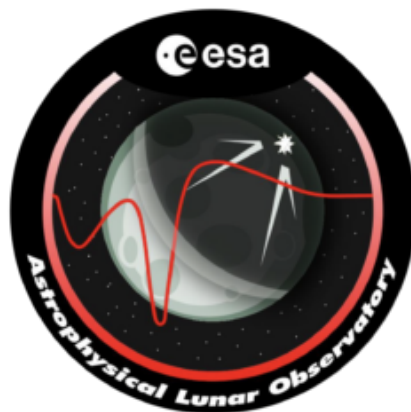
Antenna Array Geometry Reconstruction for Fast Fourier Transform Telescopes on the Lunar Far-Side

THESIS MSc PARTICLE- AND ASTROPHYSICS

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Abstract

Accurate knowledge of antenna positions is a critical requirement for Fast Fourier Transform Telescopes (FFTTs), as small deviations from a regular grid introduce phase errors that degrade imaging performance and contaminate measurements of the cosmological 21-cm signal. This work evaluates a calibration approach in which antenna geometry is reconstructed using Time of Arrival (ToA) measurements from an orbital calibrator satellite.

Simulations of a 1024 element array show that, under favorable conditions, antenna positions can be recovered with millimeter scale accuracy. Typical reconstruction errors are between 2mm and 5mm, with maxima below 1.2cm. Reconstruction performance degrades gradually with decreasing transmitter power. Horizontal position errors increase from 4.3mm ($1.4\text{m}\lambda$) at 10W to 85mm ($28\text{m}\lambda$) at 31.6mW. Increasing integration time to 0.1s improves SNR by a factor 54, enabling a proportional reduction in transmitter power, though this is limited by satellite motion during integration.

Static per-antenna delays can be calibrated within the reconstruction framework, removing systematic bias but increasing vertical position uncertainty by a factor of 3 at high power and up to a factor 15 at low power.

When compared to DEX tolerance requirements (0.027λ), the reconstruction meets accuracy requirements across a broad parameter range. These results demonstrate that orbital calibration enables accurate, in-situ reconstruction of large lunar antenna arrays without increasing surface hardware complexity.

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1 Introduction

Radio astronomy from the lunar far-side represents a unique opportunity for observational cosmology and astrophysics. Earth-based observatories struggle to detect low frequency signals due to the Earth's ionosphere and pervasive radio frequency interference (RFI) from human activity (see Figure 1). In contrast, the lunar far-side, permanently shielded from Earth's radio noise due to the Moon's tidally locked orbit, provides perhaps the best environment in the inner solar system for probing the early Universe [1]. Of particular interest is the largely unexplored period between the emission of the Cosmic Microwave Background (CMB) and the formation of the first luminous structures known as the Cosmic Dark Ages. During this epoch, the Universe was filled with neutral hydrogen gas, which emitted faint radio signals at the 21-cm hyperfine transition line [2]. Detecting this redshifted 21-cm signal offers a window into the initial conditions of cosmic structure formation and the evolution of the intergalactic medium.

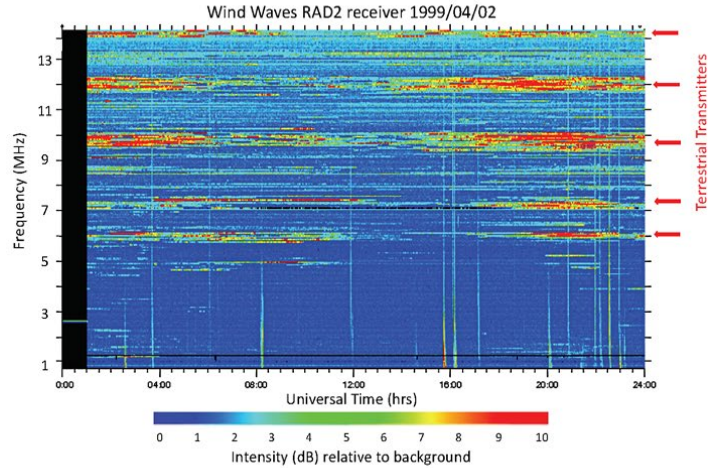


Figure 1: Observations of the low frequency radio spectrum made by the Wind/Waves spacecraft near the Moon. This shows the high level of radio frequency interference from Earth-based radio sources present even at these distances. Figure taken from J. O. Burns [1]

1.1 The 21-cm signal

The 21-cm (1.42GHz) line arises from the hyperfine splitting of the ground state of neutral hydrogen, corresponding to the energy difference between parallel and antiparallel spin alignments of the proton and electron. The relative populations of these two spin states are described by the spin temperature T_s , which determines whether the signal appears as emission or absorption relative to the CMB temperature T_{CMB} . When $T_s < T_{\text{CMB}}$, the neutral hydrogen absorbs background photons, introducing an absorption feature. When $T_s > T_{\text{CMB}}$, it emits in excess of the cosmic microwave background. The differential brightness temperature of the 21-cm signal relative to the CMB contains information about the density, thermal, and ionization state of the hydrogen gas, in a sense encoding a three-dimensional record of the evolution of the universe [3]. Its detection would bridge an observational gap in the timeline of the universe between the emission of the CMB (around 400,000 years after the Big Bang) and where galaxy surveys begin to systematically probe large-scale structure (around 1.5Gyr after the Big Bang).

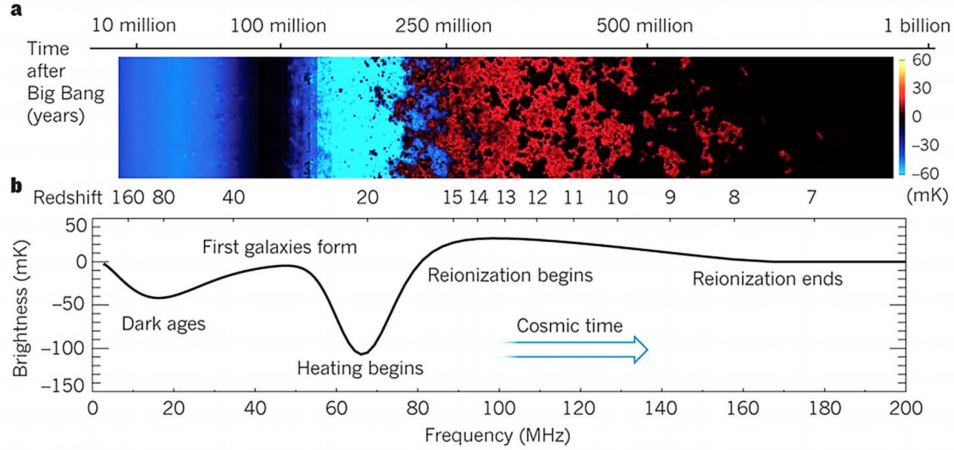


Figure 2: (a) Simulated spatial fluctuations of the redshifted 21-cm signal across cosmic time, illustrating the emergence of structure and reionization. (b) The corresponding global (sky-averaged) 21-cm brightness temperature spectrum, along with major thermal and ionization milestones such as the Dark Ages and reionization. Adapted from published simulations. Figure from Pritchard and Loeb [4].

As illustrated in Figure 2, the 21-cm signal can be viewed from two perspectives. Panel *b* shows the global (sky-averaged) brightness temperature as a function of frequency. This spectrum reflects the timing of major phases in the history of the universe, including the beginning of Lyman- α coupling, the beginning of X-ray heating, and the progression of reionization. In contrast, panel *a* depicts the spatially varying brightness field from which this average is derived. These fluctuations arise from the inhomogeneous distribution of neutral hydrogen. The global spectrum and the spatial structure taken together show that the 21-cm signal contains both an evolutionary timeline of the universe and a map of cosmic structure formation.

Despite the rich information content, these 21-cm emissions are exceedingly faint, having an amplitude four to six orders of magnitude lower than natural foreground emissions [2]. Extracting the cosmological signal from the foreground requires not only careful telescope design considerations and calibration but also state-of-the-art data analysis techniques. Small errors in the determination of the telescope’s beam pattern or frequency dependent gain can manifest as errors with a similar profile to the expected signal [5]. Mitigating these systemic errors requires precise measurements of telescope response over all observed angles and frequencies. Added to this difficulty, Earth-based observatories suffer from additional systemic errors. The ionosphere becomes increasingly refractive and absorptive below 100MHz, introducing distortions that are difficult to calibrate with the extreme precision that is required due to the large difference in flux density between the foreground and the signal of interest [6]. Man-made RFI ranging from FM broadcasts to communication satellites dominates the low-frequency spectrum even in remote locations.

1.2 Experimental efforts for 21-cm cosmology

The dual nature of the signal motivates two different experiment approaches. Global signal experiments such as EDGES[7], SARAS[8], PRIZM[9] and LEDA[10] use a small number of wide-beam antennas to measure the global spectrum shown in Figure 2b. Their goal is to detect broad absorption or emission features associated with cosmic dawn. These experiments rely on precise calibration and modeling of the spectrally

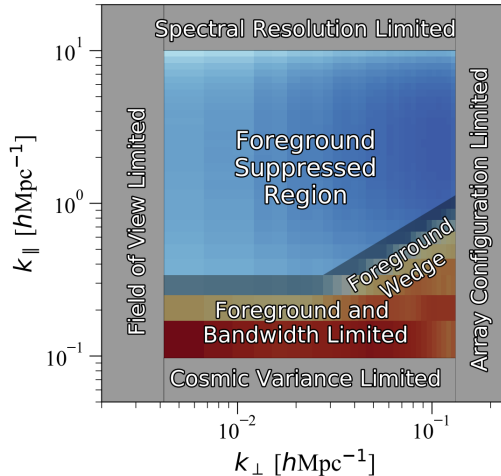


Figure 3: Accessible Fourier modes for interferometric 21-cm observations. Spectrally smooth foregrounds occupy a wedge-shaped region in $(k_{\perp}, k_{\parallel})$ space due to instrumental chromaticity, while the remaining foreground suppressed region, commonly called the EoR window, provides a comparatively clean region for detecting the cosmological signal. The extent of both regions is set by the instrument configuration and observing bandwidth. Adapted from Liu & Shaw [3]

smooth foreground to extract the weak 21-cm cosmological component from bright galactic emission [3]. The EDGES collaboration reported a detection of an absorption feature at 78MHz [11], creating a lot of excitement but also criticism due to the unexpectedly large amplitude of the measured trough. Later research questions the validity of the result, showing it can also be explained as a foreground modeling error [12].

Interferometric arrays aim to measure the spatial fluctuations of the 21-cm signal shown in Figure 2a. Interferometers such as LOFAR[13], HERA[14] and the upcoming SKA[15, 16] use many antennas to sample the Fourier modes of the sky. This enables the reconstruction of the 21-cm spatial power spectrum as a function of redshift. In Fourier space, bright foregrounds tend to occupy a distinct, spectrally smooth region referred to as the *wedge*, allowing experiments to isolate the Epoch of Reionization (EoR) window where the 21-cm signal fluctuations are detectable [17]. This is shown visually in Figure 3. These measurements probe the clustering of early galaxies, the size, and distribution of ionized regions, and the astrophysical processes driving the transition from the Dark Ages to reionization. In combination with global signal measurements, interferometric observations provide a three-dimensional view of the early universe.

The exceptional difficulty of detecting the faint 21-cm signal has led the field to search for better radio environments. This has resulted in a growing number of national and international collaborations to conceptualize and develop space- and moon-based radio observatories. The European OLFAR (Orbiting Low Frequency Antennas for Radio Astronomy) concept [18] explored an orbital interferometer architecture for observing long wavelength signals inaccessible from Earth. NASA’s FARSIDE (Farside Array for Radio Science Investigations of the Dark Ages and Exoplanets) [19] envisions a lunar surface deployed array of dipole antennas designed to detect the redshifted 21-cm signals from the early Universe, as well as low-frequency emissions from magnetized exoplanets. Another notable proposal is DARE (Dark Ages Radio Explorer) [20], a lunar orbiter intended to measure the sky-averaged 21-cm signal while periodically passing over the radio-shielded lunar far-side.

1.3 DEX and Fast Fourier Transform Telescopes

Another initiative is the Dark-ages EXplorer (DEX) [21], a mission concept spearheaded by the Astrophysical Lunar Observatory (ALO) Topical Team. DEX aims to detect and image the highly redshifted 21-cm hydrogen signal from the Dark Ages and Cosmic Dawn epochs of the Universe, corresponding to redshifts of $z = 200 - 13$. Mapping this signal over time and sky position would enable three-dimensional tomography of the early Universe, effectively capturing the evolution of cosmic structure before reionization. In order to achieve this, DEX envisions a large lunar antenna array built as a Fast Fourier Transform Telescope (FFTT), a computationally efficient radio interferometer architecture minimizing the required computational power for in-situ data processing using the Fourier relationship between the sky brightness and the measured electric fields across a planar grid of antennas [22]. FFTTs replace traditional correlators, whose computational cost scales quadratically with the number of antennas, with Fast Fourier Transform (FFT)-based image synthesis that scales as $\mathcal{O}(N \log(N))$ in the number of antennas. This enables the use of thousands of elements without prohibitive power- and computation-demands. This is of particular importance for data processing in resource-constrained environments such as the lunar surface.

A core requirement of FFTTs is the placement of antennas on a regular, cartesian grid. This geometry enables the use of FFTs which exploit this Cartesian grid symmetry for rapid image synthesis, bypassing the more computationally demanding correlation and imaging steps typical of conventional aperture synthesis techniques. The anticipated DEX deployment, however, introduces a major challenge: the automated deployment of antennas across the unprepared, rugged terrain of the lunar surface. Whether through inflating structures or by robotic means, some level of positional error is unavoidable. Deviations from the ideal grid can significantly degrade the performance of an FFTT, introducing artifacts in the reconstructed images, as can be seen in Figure 4.

1.4 Focus of this thesis

This challenge motivates the focus of this thesis: the development of a method to accurately determine the as-deployed relative antenna positions in a lunar FFTT array. Rather than requiring direct physical measurements of antenna locations, the approach pursued here infers the array geometry using calibration radio measurements. Specifically, the method exploits Time of Arrival (ToA) differences of a signal transmitted by a calibrator transmitter satellite in Low Lunar Orbit (LLO). This satellite continuously transmits a low-power, broadband pseudorandom signal across the observing band of the surface array. The signal is designed to be spectrally flat and deterministic, allowing it to be correlated with the received signals at each antenna, resulting in accurate per-antenna ToA measurements. Differences in arrival time between pairs of antennas yield Time Difference of Arrival (TDoA) measurements, which encode the projected baselines of the antenna pairs. The concept of using such an orbital transmitter to support calibration is inspired in part by the LuSEE-Night experiment [23], which employs a calibration satellite to enable precision calibration of a surface radio observatory designed for low-frequency radio measurements from the lunar far-side.

As the calibrator progresses along its orbit, it illuminates the array from a wide range of known directions. The relative positions of the antennas determine a set of baseline vectors connecting antenna pairs, and these baselines are initially unknown. Under the far-field approximation, the measured TDoAs are related to the unknown baseline vectors. Specifically, each TDoA is given by the dot product of a baseline vector with the calibrator’s line-of-sight direction. By integrating these measurements

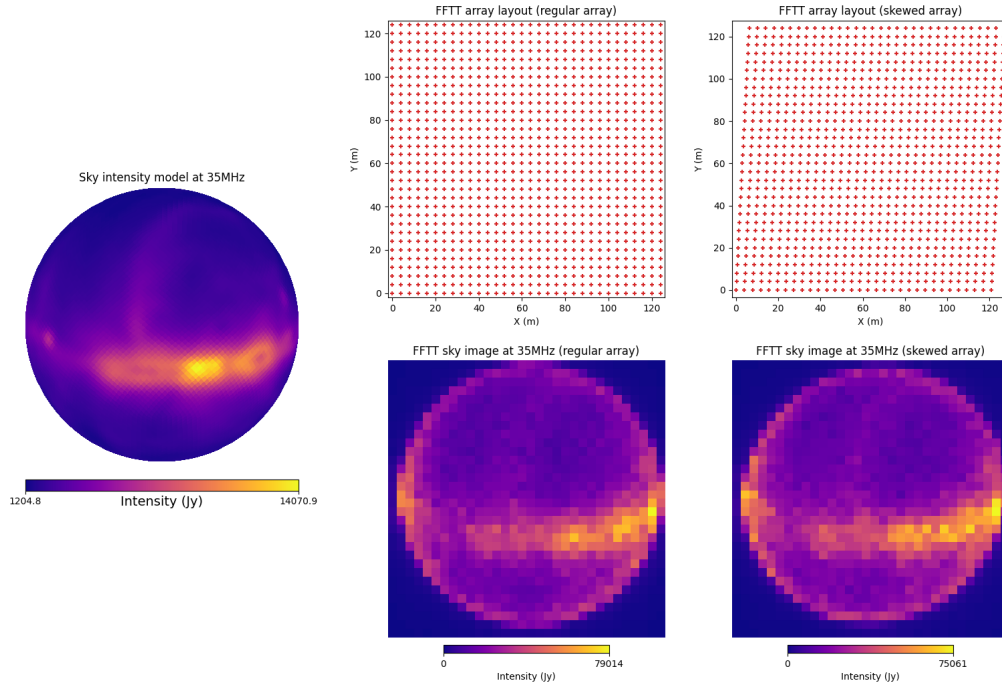


Figure 4: A demonstration of the effect of applying FFTT imaging to a skewed FFTT array. Left, a sky model that is imaged. Middle, an image generated of the sky model using simulated data with a perfectly regular array. Right, the same sky model imaged using simulated data from an array with skewed antenna positions. The resulting sky image is also distorted, as is best seen by looking at the oval-shaped horizon.

across many satellite positions, an overdetermined system of equations is built that relates the unknown baseline vectors to the known satellite directions and measured TDoAs. The baseline vectors can then be estimated by solving this system using a least-squares optimization procedure. From these baseline vectors, the geometry of the array can be inferred. With the recovered geometry information, the measured electric field data can be interpolated back onto an ideal grid (e.g., using the EPIC algorithm [24]), restoring the conditions required for FFT-based imaging.

This calibration approach is well aligned with the constraints of lunar FFTT arrays. It requires no additional surface infrastructure, integrates directly into the signal processing pipeline, and can operate at transmission powers low enough to avoid contaminating science observations. Additionally, it may be possible, provided this does not introduce strong parameter degeneracies, to use the same calibrator measurements to characterize other instrumental systematics such as per antenna beam pattern variations. Such quantities are essential for extraction of the 21-cm signal.

The remainder of this thesis is structured as follows. Chapter 2 presents the methodology, beginning with an overview of Fast Fourier Transform Telescopes and the requirements they impose on array geometry. It then details the proposed reconstruction approach based on Time Difference of Arrival measurements, including both the plane wave approximation and the refined spherical wave model, followed by a description of the simulation framework used to evaluate the method. Chapter 3 presents the results of this analysis, starting with validation under ideal conditions and then exploring the impact of practical limitations such as reduced transmitter power, integration time trade-offs, and the presence of static antenna delays. The implications of these results

for the DEX mission are also discussed. Chapter 4 discusses the assumptions made throughout the work and discusses their potential impact on real-world performance, highlighting areas for future research. Finally, Chapter 5 summarizes the main findings and conclusions.

2 Methodology

2.1 Fast Fourier Transform Telescopes (FFTTs)

The Fast Fourier Transform Telescope (FFTT) is a novel interferometer architecture that exploits a regular Cartesian grid of antenna positions to enable efficient image synthesis using the Fast Fourier Transform (FFT) algorithm. The electric field signals $E(\vec{x}_i, t)$ from antennas located on uniform Cartesian grid positions $\vec{x}_i$ are first Fourier transformed in time to obtain frequency-domain electric field samples $\tilde{E}(\vec{x}_i, f)$. Then, for each frequency channel, a spatial 2D FFT is applied across the grid:

$$I(\vec{u}, f) = F_{2D}[\tilde{E}(\vec{x}, f)] \quad (1)$$

where:

- $\vec{u} = \frac{\vec{x}}{\lambda}$ is a spatial frequency coordinate in units of wavelength;
- $F_{2D}[\cdot]$ is a two dimensional spatial Fourier transform;
- I is a measure of sky brightness.

This procedure yields a dirty image (an image convolved with the interferometer's beam pattern) of the sky brightness distribution in $\mathcal{O}(N \log(N))$ computational time, where N represents the number of antennas in the interferometer.

Any deviation from a perfect grid breaks the assumptions of the FFT-based image generation. These deviations introduce phase errors in the spatial transform, which manifest as image artifacts. This implies FFTTs require pre-processing corrections to restore grid regularity before imaging. Therefore, an essential challenge for FFTT deployment on the rugged lunar surface is accurately determining the true, as-deployed positions of the antenna elements. Once these positions are known, the measured electric field values can be interpolated back onto an ideal grid (e.g., using the EPIC algorithm [24]), restoring the conditions necessary for FFT based image reconstruction.

2.2 Determination of array geometry

In order to enable FFT-based imaging with an imperfectly placed antenna array, the actual antenna positions after deployment must be determined. This can be achieved by fitting the relative geometry of the array using measured arrival times of a signal from a known calibrator source in Low Lunar Orbit (LLO).

The geometry reconstruction is performed in two passes. Firstly, by making use of a plane wave approximation. Under this assumption, the problem reduces to a linear least squares problem, which can be solved efficiently, but it introduces a small bias. The solution is refined using a spherical wave model, removing bias at the cost of longer compute times. The full method is described below.

2.2.1 Measuring Time Difference of Arrival (TDoA)

To determine the antenna array geometry, the ToA of a known reference signal at each antenna needs to be measured first. This is done by correlating the received antenna signals with the known waveform transmitted by a calibrator. Assume the calibrator transmits a periodic deterministic pseudorandom wideband signal $s(t)$ with period T . The signal at antenna i is given by:

$$r_i(t) = G_i \cdot s(t - \tau_i) + n_i(t) \quad (2)$$

where:

- τ_i is the geometric signal propagation delay between the calibrator and antenna i ;
- G_i is the gain of antenna i , depending on the direction of arrival of the signal;
- $n_i(t)$ is the additional noise received by antenna i .

Since $s(t)$ is known, one can recover τ_i by correlating $r_i(t)$ against $s(t)$ and locating the peak. This is done through a numerical cross-correlation between one period of the calibrator waveform $s(t)$ and the sampled signal $r_i(t)$. Quadratic interpolation around the bin with the highest peak and its two neighbors gives a measurement of the ToA $\hat{\tau}_i$ modulo the calibrator signal period T and up to some global time offset. If the period T is much greater than the maximum geometric time delay between any pair of antennas, the TDoA can be uniquely estimated by taking the smallest signed difference between two antennas ToA modulo T :

$$\hat{\Delta}\tau_{ij} = \left(\left(\hat{\tau}_i - \hat{\tau}_j + \frac{T}{2} \right) \bmod T \right) - \frac{T}{2} \quad (3)$$

2.2.2 Estimating baseline vectors

Let $\vec{x}_i \in \mathbb{R}^3$ be the unknown position vector of antenna i . A calibrator in LLO transmits a known, broadband pseudo-random signal $s(t)$, which is received as in equation 2. Assuming perfect vacuum conditions, the geometric signal propagation delay is given by:

$$\tau_i = \frac{1}{c} \|\vec{x}_i - \vec{x}_s\| \quad (4)$$

where $\vec{x}_s \in \mathbb{R}^3$ is the position of the calibrator at the time of signal transmission.

Plane wave approximation

Since the distance between the array and the calibrator is much greater than the distance between any two elements in the array, a far-field approximation can be used, simplifying the incoming wavefront to a plane wave. Under this approximation, we can simplify τ_i :

$$\tau_i \approx \tau_0 - \frac{1}{c} \vec{s} \cdot \vec{x}_i \quad (5)$$

where:

- $\vec{s}$ is the unit direction vector of the plane wave;
- τ_0 is a common delay term across all antennas.

By taking the TDoA between two antennas i and j , τ_0 is eliminated, and we find:

$$\Delta\tau_{ij} = \tau_i - \tau_j \approx -\frac{1}{c} \vec{s} \cdot \vec{b}_{ij} \quad (6)$$

where $\vec{b}_{ij} = \vec{x}_i - \vec{x}_j$ is the baseline vector between antennas i and j .

As the calibrator satellite progresses along its orbit, it illuminates the array from a wide variety of sky directions. Assume we have K calibrator directions $\vec{s}_1, \dots, \vec{s}_K$. For each calibrator direction, we have a measurement of the TDoA $\Delta\tau_{ij}$. Let $\mathbf{S} \in \mathbb{R}^{K \times 3}$ be a matrix with, in each row, i the calibrator direction vector $\vec{s}_i$, and $\vec{\Delta\tau}_{ij} \in \mathbb{R}^K$ be a vector of the corresponding TDoA measurements. We obtain a system of linear equations:

$$\mathbf{S} \vec{b}_{ij} \approx -c \vec{\Delta\tau}_{ij} \quad (7)$$

which can be solved in the least-squares sense:

$$\hat{\vec{b}}_{ij} = -c(\mathbf{S}^T \mathbf{S})^{-1} \mathbf{S}^T \vec{\Delta\tau}_{ij} \quad (8)$$

to recover an estimate of the baseline $\vec{b}_{ij}$. In practice, the least-squares problem is solved numerically using a Singular Value Decomposition (SVD)-based algorithm.

Spherical wave model

The far-field approximation introduces a small systematic error. For a typical calibrator distance of 300km and a baseline length of 100m, the bias due to wavefront curvature can be on the order of 1.7cm. While small, this bias is worth eliminating because it introduces spatially correlated antenna position errors. Such correlated geometric errors are particularly problematic, as they produce baseline-dependent spatially correlated phase offsets, which are more detrimental for 21-cm cosmology measurements than stochastic random perturbations [25]. To mitigate this, the initial geometry estimate is used to refine the baseline vectors by directly fitting the exact spherical wave model.

For a given antenna pair (i, j) , we consider the exact difference in propagation distances:

$$c \Delta\tau_{ij} = \|\vec{x}_i - \vec{x}_s\| - \|\vec{x}_j - \vec{x}_s\|. \quad (9)$$

Using the initial geometry estimate, antenna i is taken as a reference, and the baseline $\vec{b}_{ij}$ is re-estimated by minimizing the residual between the measured and modeled TDoA over all calibrator positions:

$$\mathcal{R}(\vec{b}_{ij}) = \sum_k \left[\|\vec{x}_s^{(k)} - \vec{x}_i\| - \|\vec{x}_s^{(k)} - (\vec{x}_i + \vec{b}_{ij})\| - c \hat{\Delta\tau}_{ij}^{(k)} \right]^2. \quad (10)$$

This nonlinear least-squares problem is solved iteratively using the Levenberg-Marquardt algorithm, initialized with the far-field estimate $\hat{\vec{b}}_{ij}^{(0)}$. The resulting refined baselines $\hat{\vec{b}}_{ij}^{(1)}$ account for wavefront curvature and are therefore free of model bias.

Conditioning

The stability of the baseline estimation problem is governed by the conditioning of the matrix $\mathbf{S}$, which encodes the calibrator direction vectors. The least-squares solution involves the matrix $(\mathbf{S}^T \mathbf{S})^{-1}$ whose sensitivity to noise depends on its condition number. Each TDoA measurement constrains the projection of a baseline vector onto the calibrator line of sight. A well-conditioned reconstruction therefore requires a set of calibrator directions that span the sky sufficiently, such that all components of the baseline vectors are independently constrained.

2.2.3 Array geometry reconstruction

The baseline vectors $\vec{b}_{ij}$ are differences of antenna positions. To reconstruct the geometry $\{\vec{x}_i\}$ up to a global translation, the position of an arbitrary reference antenna can be fixed (say, $\vec{x}_1 = \vec{0}$). An optimization problem that minimizes the difference between estimated and reconstructed baselines can be defined using the loss function $\mathcal{L}$:

$$\mathcal{L}(\vec{x}_1, \dots, \vec{x}_N) = \sum_{i < j} w_{ij} \|\vec{x}_i - \vec{x}_j - \hat{\vec{b}}_{ij}\|^2 \quad (11)$$

where w_{ij} is a weight for the contribution of the baseline measurement $\hat{\vec{b}}_{ij}$ to the loss function, which can, for example, be derived from the residuals of the least squares fit

of the baseline vector. Minimizing $\mathcal{L}$ over $\vec{x}_2, \dots, \vec{x}_N$ with $\vec{x}_1 = \vec{0}$ solves the geometry of the array that best fits the baseline measurements. This minimization problem is effectively solved using the LSQR algorithm, which is particularly well suited to such large and sparse problems.

2.3 Simulation framework for geometry reconstruction

To evaluate the performance of the proposed geometry reconstruction technique, a complete end-to-end simulation was developed. This simulation models a radio interferometric array, background sky signals, instrumental noise, and a calibrator source in LLO.

The simulation framework is designed to model the full broadband time domain E-field measurements of the array rather than interferometric visibilities. The simulation framework works in discrete steps, where one step produces an integration window consisting of a batch of time domain samples from each antenna. Within such an integration window, all geometry (the locations of all background sources and the calibrator) is assumed stationary. For each integration window, per antenna in the array, a high resolution frequency domain spectrum is synthesized with contributions from the diffuse galactic foreground emission, instrumental noise, and calibrator signal. This spectrum is then inverse Fourier transformed into a set of time domain E-field samples. These samples are then processed to extract Time of Arrival measurements of the calibrator signal.

The various components of the simulation pipeline are explained in more detail below.

2.3.1 The interferometric array

The simulated antenna array consists of a two dimensional square Cartesian grid of 32×32 cross-dipole antennas, for a total of 1024 elements. Each antenna is modeled to have a 4m diameter, and the grid has 4m spacing between antenna centers such that the cross dipoles are densely packed. For the moment, the near-field effects of this dense packing on the antenna beam shapes are ignored. The layout can be seen in Figure 5 (left panel).

Each antenna receives signals over a passband of 50–100MHz, representing a 50MHz observing bandwidth centered around 75MHz. This signal is sampled at 120MHz after downmixing by 45MHz to center the observing band in the 0 – 60MHz bandwidth obtained from the sample frequency with a 5MHz guard band on each side.

The gain of each element is modeled as:

$$G(\theta) = \cos^2(\theta), \quad (12)$$

where θ is the angle between incoming radiation and the zenith of the array. A visualization of this gain model can be seen in Figure 5 (middle panel). This is a typical gain model for cross-dipole antennas. Antenna and signal polarizations are not yet considered in this analysis. The effective aperture of each element is modeled as the minimum of the physical aperture and the theoretical effective aperture:

$$A_e = \min\left\{A_p, \frac{\lambda^2}{4}\right\} = \min\left\{4, \frac{\lambda^2}{4}\right\}, \quad (13)$$

where λ is the wavelength of incident radiation. The effective aperture as a function of frequency is shown in Figure 5 (right panel).

The antenna noise temperature is divided into two contributions: the sky noise temperature T_{sky} representing signal contribution from galactic radio emissions, and

the system noise temperature T_{sys} representing noise contribution from the receiver chain and thermal noise. The antennas are taken to be sky noise limited, making T_{sky} the dominant contribution to the total noise temperature. T_{sky} is modeled by simulating the galactic radio sky as point sources, discussed in more detail in the next chapter. T_{sys} is given the constant value of 5000K over the observing band, which is slightly higher than the expected figure for the lowest noise receiver front-end investigated by J.C.F. Zandboer's Master's thesis on antenna design for lunar detection of global 21-cm signals [26], as can be seen in Figure 6.

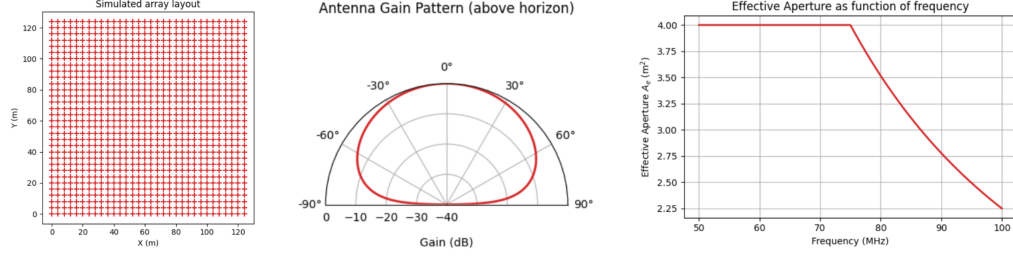


Figure 5: A visualization of the simulated array layout (left), the antenna beam pattern (middle), and the antenna effective aperture (right).

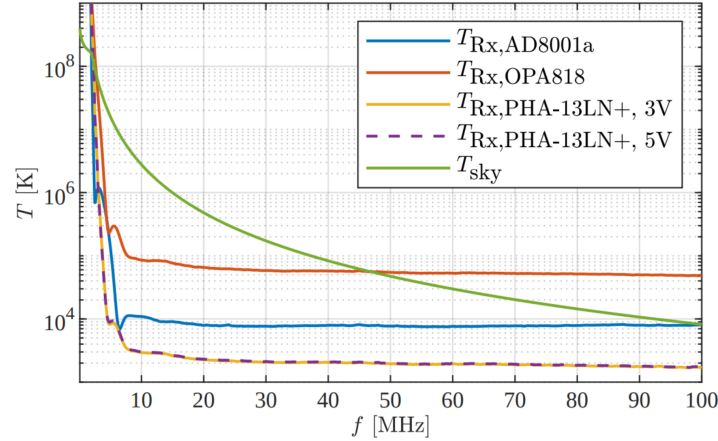


Figure 6: Simulated equivalent isotropic receiver noise temperature T_{RX} of an active antenna system as a function of frequency, compared against the expected sky noise temperature T_{sky} . The results illustrate that, across the observing band, the lowest noise receiver frontend has an approximately constant noise temperature of around 1000K. Adapted from J.C.F. Zandboer, *Active Antenna Design for Lunar Detection of Global 21 cm-signals from the Dark Ages*, MSc thesis, Eindhoven University of Technology, 2023 [26]

2.3.2 Background sky signal

The diffuse galactic radio emission represents the majority contribution to the system noise temperature of the array; thus, a realistic model of this emission is required. To simulate this emission, a point source sky model is used, where each source is defined by a sky location (right ascension and declination), intensity I and spectral index α . The intensity of each source as a function of frequency is given by:

$$I(f) = I_0 \left(\frac{f}{f_0} \right)^\alpha \quad (14)$$

where I_0 is the source's intensity at reference frequency f_0 .

To decompose the diffuse galactic radio emission into this point source model, data from the global sky model GSM2016 was used via the PyGDSM package [27]. While PyGDSM provides models from a variety of surveys, the GSM2016 survey was chosen as it provides sky maps in physical units (MJy/sr) instead of antenna specific units (e.g., antenna temperature). The GSM2016 all-sky radio maps are based on a combination of surveys over a range of frequencies and angular resolutions [28]. PyGDSM provides sky maps from this survey at all frequencies in the observed band of the array.

The GSM2016 maps use the HEALPix tessellation model, in which the sky is divided into pixels of equal area. The parameter n_{side} controls the angular resolution of the map, with the total number of pixels given by $n_{\text{pix}} = 12n_{\text{side}}^2$. For $n_{\text{side}} = 32$, this results in 12,288 sky pixels with an angular scale of approximately 1.8° . This resolution was chosen as a compromise between computational cost and sufficient representation of galactic diffuse radio structure. Since the purpose of the sky model in this work is primarily to provide realistic foreground noise, higher angular resolution is not required.

A reference skymap with resolution $n_{\text{side}} = 32$ at a frequency f_0 near the center of the observing band is first generated. Each pixel in this map is converted into a point source, defining the sky source positions and intensities I_0 .

To determine the spectral index of each source, additional sky maps are generated at 10 evenly spaced frequencies across the observing band. A log linear fit of these skymaps is done, making use of the relation:

$$\log(I(f)) = \log(I_0) + \alpha \log\left(\frac{f}{f_0}\right), \quad (15)$$

such that the spectral index α is the slope of the fit. Figure 7 shows the intensity I_0 and spectral index α of each point source.

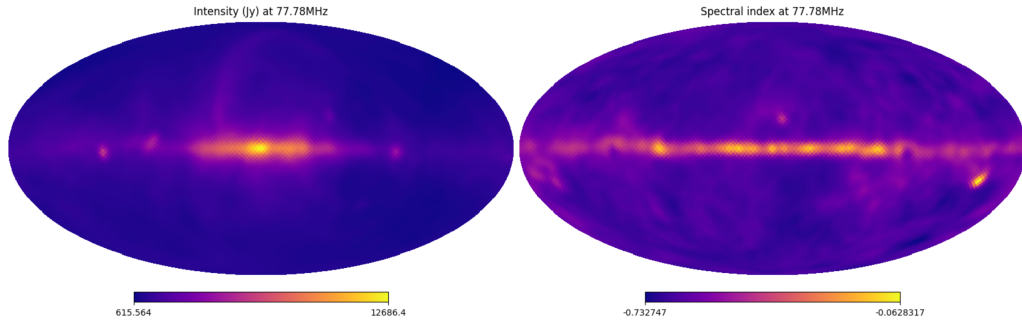


Figure 7: The intensity I_0 and spectral index α of all point sources modelling the diffuse galactic radio emission.

These sky sources are simulated as far-field plane waves at many discrete frequencies, with direction derived from the sources sky coordinates. The electric field contribution of each source at an antenna is modulated by the antenna's direction dependent gain model. This process is discussed in more detail in Chapter 2.3.4.

2.3.3 The calibrator

A simulated calibrator is placed in a low lunar orbit. For the orbit, a 75° inclined circular orbit with a mean altitude of 100km above the lunar surface is used. These orbital parameters provide good sky coverage, and the low altitude allows for low transmission power. A week of this orbit is simulated using the TudatPy astrodynamics library, with

a resolution of one position per second. Since the objective here is merely to obtain a realistic distribution of calibrator positions on the sky, the exact orbital dynamics are not critical. Consequently, the orbit can be propagated using a standard TudatPy configuration, including point-mass Moon, Earth and Sun gravity models and Runge-Kutta fixed step size integrator. The resulting sections of the orbit visible to the simulated array can be seen in Figure 8.

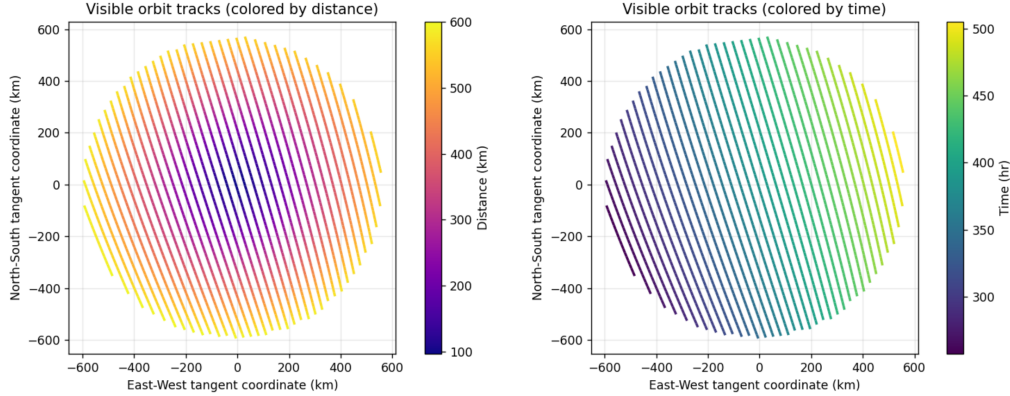


Figure 8: A visualization of the section of simulated calibrator orbit visible to the array. Left: the orbital passes above the array’s local horizon, colored by the Euclidian distance between the calibrator satellite and the array. Right: the orbital passes above the array’s local horizon, colored by orbital time.

The calibrator broadcasts with a uniform power over the observing band of the array. A pseudorandom periodic waveform is generated in the downmixed frequency band (5 – 55MHz) and is upmixed by 45MHz before transmission. In this way, after downmixing and sampling on the receiver end, a discretization of the original pseudorandom signal is recovered, preserving its shape and periodicity. To ensure the simulated signal has a uniform power spectrum and is periodic, an Inverse Fast Fourier Transform (IFFT) is used to create it. In this work, a calibrator signal with a period $T = 137\mu\text{s}$ is used, generated from an IFFT of a complex frequency spectrum containing 16384 bins with a frequency spacing of 7.32KHz. For each frequency in the discrete frequency spectrum, a complex number is generated with a uniformly random phase and a fixed magnitude of either a constant factor derived from transmitted power if the frequency is in the 5 – 55MHz band, or 0 otherwise. This spectrum is then processed with an IFFT, resulting in a time-domain signal with the desired properties. During frequency spectrum synthesis, positive and negative frequencies are treated independently, meaning the resulting time-domain signal is complex-valued. The $137\mu\text{s}$ period would allow for non-ambiguous TDoA extraction for baselines up to 20km in length.

2.3.4 Simulation pipeline

To generate data, a simulation pipeline is needed which supports simulating measurement data of the array model incorporating both the custom diffuse sky model and the calibrator source.

The use of an industry-standard radio-interferometric array simulation tool, namely SAGECAL[29] was investigated. The problem with using SAGECAL for this research is the data product it produces is incompatible with the research done in this thesis. This is discussed further in Appendix A.

Custom simulation

To solve the shortcomings of **SAGECAL** for this purpose, a custom simulation pipeline was developed from the ground up. It explicitly avoids visibility synthesis, instead producing either complex-valued electric field samples or frequency spectra on a per-antenna basis over a specified number of time steps.

The simulation pipeline starts by defining the array model, point source sky model, and calibrator as discussed in the previous sections. Of all calibrator positions in the simulated orbit, a subset is selected (e.g., all points more than 10° above the array’s local horizon) to save on data volume and not waste compute time performing simulations when the calibrator is not in view of the array. The number of selected calibrator positions defines the number of simulation steps. The integration time (the size of the integration window for each simulation step) of the receivers in the array is defined, determining the frequency resolution. The power transmitted by the calibrator is also specified, which together with the propagation distance and receiver gain determines the resulting received calibrator signal power. To save on computation time, the integration windows of the receivers are set quite short (4096 samples, or $34.1\mu\text{s}$ at 120MHz), and to compensate, the transmitter power of the calibrator is set quite high (e.g. 10W). A discussion on the trade-off between integration time and calibrator power and what real calibrator power would provide similar performance to the simulated calibrator power accounting for realistic integration time can be found in Section 3.3.

An important effect to capture in the simulation is that of real sources having a continuous frequency spectrum. When these sources are sampled discretely, their signal is captured with finite frequency resolution. As a result, in the frequency domain, each frequency bin of the captured spectrum contains the complex interference of a small range of frequencies. To simulate this effect, the simulation defines a frequency resolution setting. The simulation synthesizes frequency-domain electric field spectra such that the IFFT produces more time-domain samples than fit in the configured integration window by a factor of the configured frequency resolution. All generated time-domain samples that fall outside the integration window are discarded. In this way, when the time-domain samples are transformed back to frequency domain, multiple frequency components of the signal squeeze into the same bin of the frequency spectrum, capturing this interference effect. How this works is demonstrated in Figure 9.

A graphical overview of the simulation pipeline can be seen in Figure 10. For each step of the simulation, the next calibrator position is selected. The orbital time passed between the previous step and the new step is calculated. This time difference is used to rotate the sky model with a rotation speed matching that of a lunar sidereal month, simulating the effect of the moon rotating around its axis. A time delay is also added to the calibrator signal. All sources in the sky model are assigned uniformly random phases for each frequency in the simulated spectra to emulate incoherence and spatially distinct emission. For every receiver in the array, the following is done to generate the time-domain E-field samples:

- A discrete complex frequency spectrum is initialized to zero;
- For all frequencies in the observing band, system noise is added in the form of a complex number with uniformly random phase and magnitude derived from the system noise temperature and inversely proportional to the receiver’s effective aperture A_e ;
- For every source in the sky model and the calibrator:
 - The geometric time delay δt of the source’s signal is calculated from the dot product between the source’s direction and the receiver’s position vectors, or

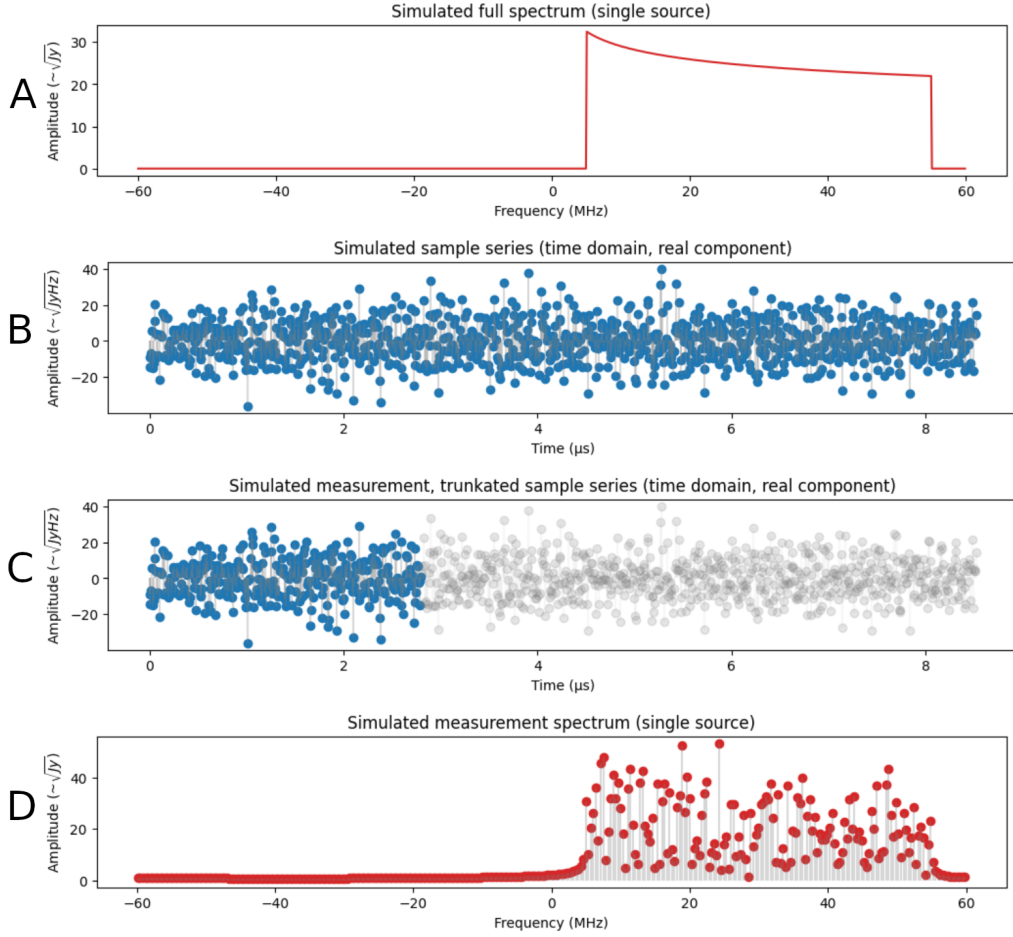


Figure 9: A demonstration of how the simulation models complex interference of different frequency components within a spectrum bin. Panel A shows a high-resolution simulated spectrum of a source without frequency interference, which is generated directly from the source's intensity and spectral index. Panel B shows the high-resolution spectrum converted to the time domain. Panel C shows the generated time domain signal truncated to the size of an integration window with the rest of the samples discarded. This is the simulated measurement. Panel D shows the frequency domain spectrum obtained by applying an FFT to the simulated measurement. This spectrum is lower resolution than the one from panel A, and each bin contains the complex interference of a number of closely spaced frequency components of the measurement signal.

using equation 4 for the calibrator, and all its phase components are incremented by a factor $2\pi f\delta t$;

- The gain $G(\theta)$ is calculated from the source’s direction or the calibrator’s relative position;
- For each frequency component of the spectrum, the source’s intensity $I(f)$ is calculated and multiplied by the gain. A complex number with the time-delayed phase and intensity equal to the square root of the calculated intensity is added to the frequency spectrum;
- The frequency spectrum is converted to time-domain using an IFFT;
- A window of these samples with size defined by the integration time is selected;

A dedicated high-performance backend was developed to enable fast experimentation and iteration. The signal synthesis from the given array-, sky- and calibrator-models is implemented as a standalone library written in Rust and WGS. It also includes Python bindings for simple pipeline integration. Two execution runtimes are implemented, a multithreaded CPU implementation and a GPU implementation.

The CPU runtime parallelizes over the receivers in the array, distributing the signal synthesis of each receiver over a configurable number of threads. Within a single antenna, the computation proceeds sequentially over all sources and frequency bins. This design avoids synchronization overhead but limits the available parallelism to the number of antennas. This approach works well for the large 1024-element array considered in this work. All calculations are done with double precision (64-bit) floats to avoid excessive rounding errors in accumulation loops.

The GPU runtime uses a finer parallelization strategy. The synthesis of electric field spectra is parallelized over frequency bins and antennas simultaneously. Each thread invocation is responsible for accumulating contributions to a single frequency bin of a single antenna. Within a thread, source contributions are processed sequentially to avoid synchronization in accumulation. As WGS does not support double precision floats, regular 32-bit floats are used. Rounding errors when accumulating are significantly reduced by using Kahan summation. The effective difference between results from the CPU and GPU runtimes is negligible.

On a system equipped with an AMD Ryzen 7 2700 CPU (16 threads) and a Radeon RX 5700 GPU, where the CPU implementation requires approximately 270 seconds to simulate one timestep, the GPU implementation completes the same workload in approximately 2.2 seconds. This corresponds to a speedup of two orders of magnitude. These performance characteristics further motivate the choice to synthesize signals in the frequency domain followed by an IFFT to obtain time domain samples. While a direct time domain synthesis approach would have a lower runtime complexity by a factor $\log(\text{\#samples})$, it would require accumulating contributions from all sources and all frequency components for each individual time sample. Such a formulation introduces two nested accumulation dimensions (sources and frequencies) per sample, making efficient GPU parallelization more difficult.

From the generated time domain sample windows, the signal ToA is extracted using correlation. Due to how the simulation generates the time domain signals, they exhibit a structural property that enables efficient correlation. The construction of the signals from frequency spectra and IFFT means the resulting time domain signals are periodic when zero-padded to the length of the calibrator signal. As a consequence, cross-correlation between the generated signals and the calibrator signal can be computed efficiently using the convolution theorem. Rather than performing direct time

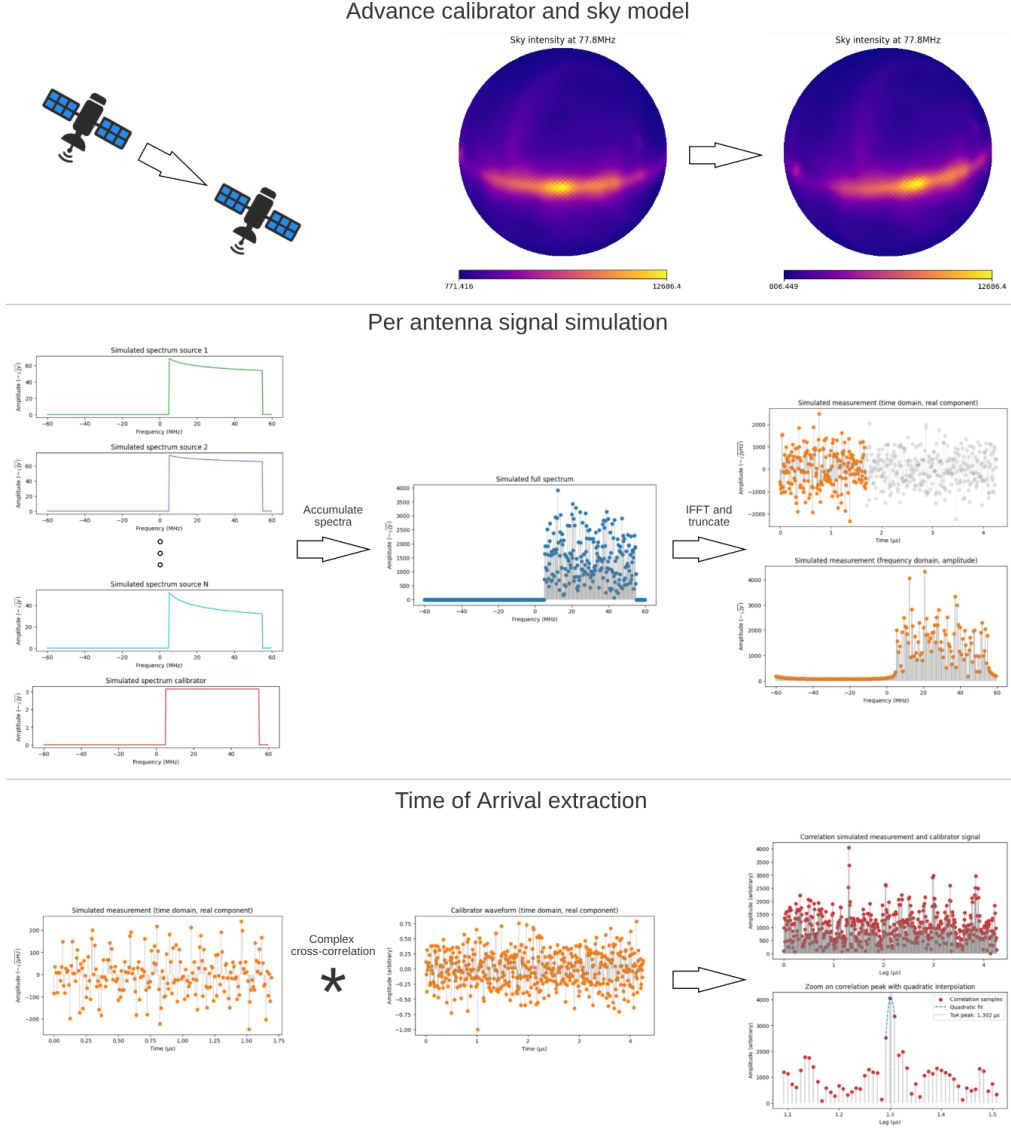


Figure 10: Schematic overview of the simulation pipeline and time of arrival extraction. First, the calibrator position is updated and the sky model is rotated. Simulated contributions from all sky sources, together with the calibrator signal, are combined to produce per-antenna time-domain measurements. The arrival time is subsequently extracted by cross-correlating the measurement with the calibrator waveform.

domain correlation, which has computational complexity $\mathcal{O}(n^2)$ where n is the number of samples, the zero-padded signals are transformed to frequency domain, where correlation reduces to complex multiplication of one spectrum with the complex conjugate of the other. An inverse Fourier transform then yields the correlation function in the time domain. This reduces the computational complexity to $\mathcal{O}(n \log(n))$.

Finally, the generated ToA data is saved, such that it can later be used as input to the geometry reconstruction algorithm discussed in Chapter 2.2.

The simulation program is published on GitHub under the name *fringe* [30].

Verification

The custom simulation pipeline was tested by comparing the data output to that of SAGECAL. The time domain electric field samples produced by the custom simulation can be converted to visibilities, allowing analysis identical to SAGECAL data. This comparison has been performed for a variety of different test sky models and can be found in Appendix B.

3 Results

3.1 Validation under ideal conditions

In this subsection, the proposed technique for inferring array geometry is validated under idealized conditions. A high-power satellite signal is assumed, together with dense and well-distributed calibrator sky coverage. These assumptions are chosen to isolate the intrinsic performance of the inference method, independent of practical limitations such as low SNR or unfavorable calibrator sky coverage.

As the true array geometry is known, results can be directly compared to estimated quantities at each stage of the processing chain. The results are presented progressively, following the sequence of operations described in Chapter 2.2.

3.1.1 Simulation settings

The array model described in Section 2.3.1 is used. Per calibrator position, a sample window of 4096 samples is generated, corresponding to an integration time of $34.1\mu\text{s}$. For the calibrator, one full set of orbital passes covering the area of the sky more than 10 degrees above the array's local horizon is included, with a density of one position per second for a total of 7166 positions. The calibrator's pseudorandom signal has period $T = 137\mu\text{s}$, which is more than sufficient to uniquely determine time differences of arrival.

3.1.2 Time of Arrival estimation

Time of Arrival (ToA) estimates are obtained for each array element by correlating the received signal with the known satellite waveform, followed by quadratic interpolation around the correlation peak (Section 2.2.1). The ToA estimation error is defined as

$$\Delta\tau = \tau_{\text{est}} - \tau_{\text{true}}, \quad (16)$$

where τ_{true} is computed from the known geometry and satellite orbit.

Figure 11 shows a map of the ToA errors across all array elements and observations.

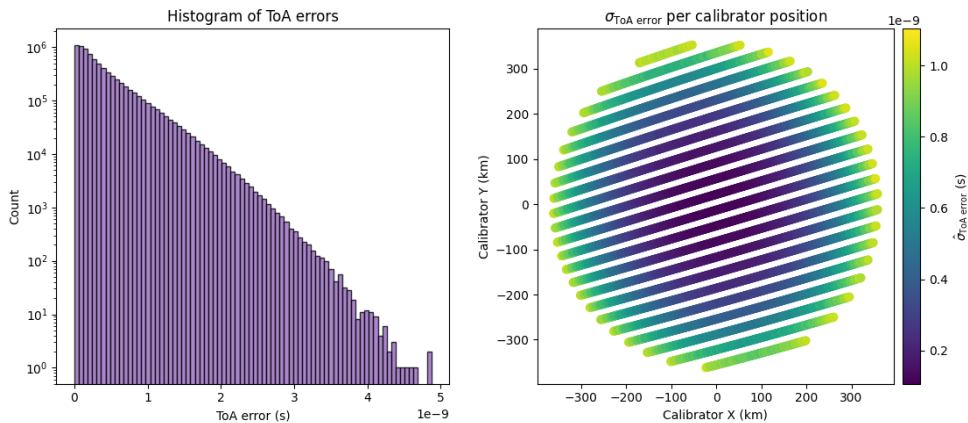


Figure 11: Distribution and spatial structure of time-of-arrival (ToA) errors. Log-scale histogram of ToA errors (left), spatial map of the standard deviation of ToA error ($\sigma_{\text{ToA error}}$) across calibrator positions (right).

The various effects that change the distribution from one calibrator position to the next combine to give a total error distribution that is close to exponential. The majority of ToA measurements are sub-nanosecond accurate.

For a fixed calibrator position, the ToA errors are Gaussian distributed. The standard deviation depends strongly on calibrator position. For positions closer to the horizon, not only is the calibrator further away from the array, but the gain of the receivers is also poor, leading to distributions with larger spread.

3.1.3 Array Geometry Reconstruction

Plane-wave approximation solution

The array geometry is reconstructed following the procedure outlined in Section 2.2. Time differences of arrival are first converted to baseline estimates using the plane-wave approximation, in which the incoming wavefront is assumed to be planar and characterized by a direction vector $\vec{s}$. For each antenna pair, a linear system is formed over all calibrator directions and solved in a least-squares sense to obtain an estimate of the baseline vector. The full array geometry is then recovered by minimizing the global consistency loss function over all baselines, with one antenna position fixed.

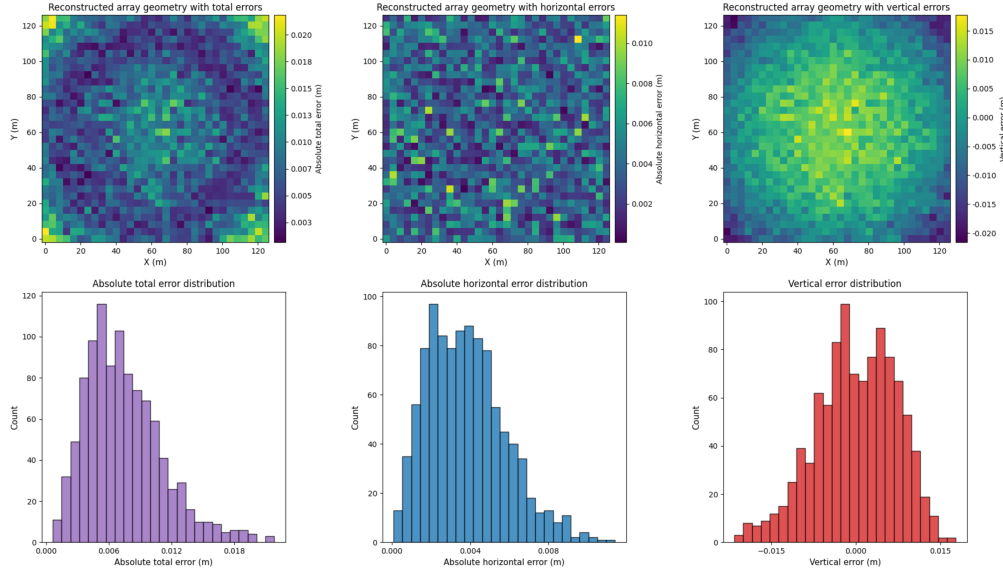


Figure 12: Reconstructed array geometry with plane wave approximation. Top row: spatial maps of total position error (left), horizontal error magnitude (center), and vertical error (right). A clear position-dependent bias is observed in the vertical component, with a radial structure across the array. Bottom row: corresponding error distributions. The horizontal errors are approximately Rayleigh distributed, consistent with isotropic noise, while the vertical errors have a multimodal non-Gaussian distribution, indicating systematic errors.

The results in Figure 12 show that, under the idealized measurement conditions, the plane wave approximation introduces a visible systematic bias in the reconstructed geometry. This bias is most pronounced in the vertical component, where a clear spatial structure is visible across the array. The horizontal components of the baselines are observed symmetrically from all directions, resulting in the bias from the plane wave approximation cancelling out. The vertical components of the baselines are exclusively observed from above the array, meaning the bias is retained.

These results demonstrate that, under ideal conditions, the accuracy of the reconstructed geometry using this technique is limited by the plane wave approximation, motivating the use of a more accurate spherical wavefront model.

Spherical wavefront solution

To remove the bias introduced by the plane-wave approximation, the reconstruction is repeated using the exact spherical wavefront model described in Section 2.2. In this formulation, the full geometric delay expression is retained, and time differences of arrival are directly related to antenna positions through the nonlinear dependence on the calibrator location. This eliminates the far-field approximation and correctly accounts for wavefront curvature.

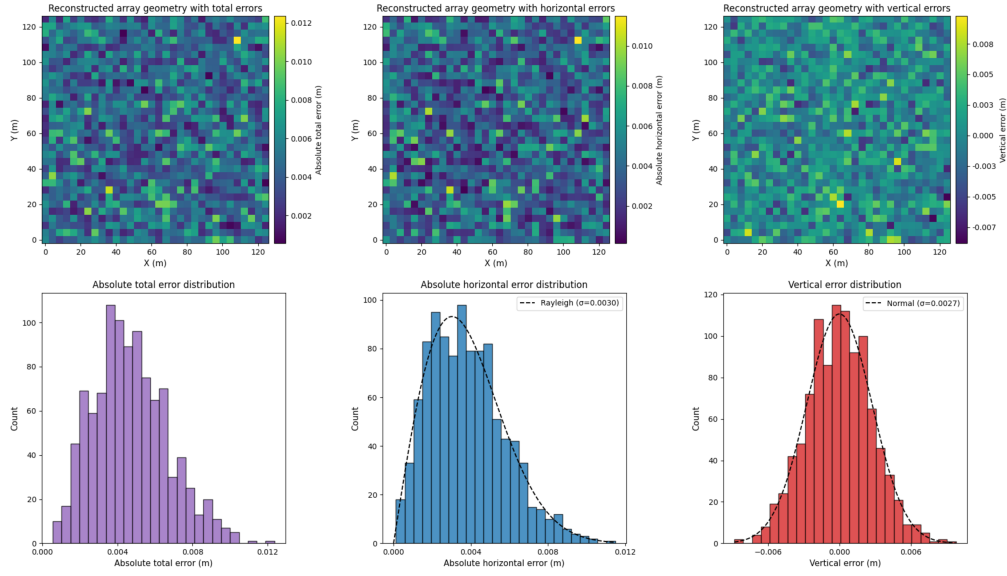


Figure 13: Reconstructed array geometry with plane wave approximation. Top row: spatial maps of total position error (left), horizontal error magnitude (center), and vertical error (right). The vertical bias seen under the plane wave approximation is no longer present, and errors appear spatially uncorrelated. Bottom row: corresponding error distributions. The horizontal errors are approximately Rayleigh distributed, and the vertical errors are approximately Gaussian distributed, indicating errors are dominated by stochastic noise rather than modeling errors.

The results in Figure 13 confirm that the systematic bias observed under the plane-wave approximation is effectively removed when using the spherical wavefront model. The spatial error maps show no residual structure, and the vertical error component is centered around zero across the entire array. Furthermore, the error distributions are consistent with purely stochastic measurement noise: the horizontal components follow a Rayleigh distribution, while the vertical component is well described by a Gaussian distribution.

Quantitatively, the reconstruction accuracy improves significantly. The total position error is reduced to the millimeter level, with typical errors on the order of a few millimeters and maxima below approximately 12mm. Horizontal errors are slightly larger than vertical errors. When the calibrator is near zenith, it is relatively close to the array and observed with high gain, yielding high-precision delay estimates that strongly constrain vertical components of baselines. In contrast, when the calibrator is near the horizon, it is both more distant and observed with reduced antenna gain, lead-

ing to noisier measurements that contribute less effectively to constraining horizontal geometry.

These results demonstrate that, with the correct physical model, the proposed method is capable of recovering the full array geometry with high accuracy and without bias.

3.1.4 Summary

Under idealized conditions of high signal-to-noise ratio and favorable calibrator sky coverage, the proposed technique accurately recovers Time of Arrival measurements, baseline vectors, and the full array geometry. Errors at each stage are small, unbiased, and consistent with theoretical expectations based on measurement noise and system conditioning. This establishes a performance baseline against which more realistic scenarios are evaluated in the following sections.

3.2 Sensitivity to transmitter power

In this subsection, the dependence of array reconstruction accuracy on the calibrator transmitter power is investigated. The analysis follows the same procedure as described in Section 3.1, but is repeated for a range of transmitter powers below the 10 W used in the ideal case. Specifically, ten additional power levels are considered, logarithmically spaced according to

$$P_i = 10 \cdot 10^{-i/4}, \quad i = 1, \dots, 10, \quad (17)$$

resulting in a gradual reduction in Signal to Noise Ratio (SNR) over approximately 2.5 orders of magnitude. How these power levels compare to both the galactic foreground model and the system noise contributions can be seen in Figure 14. These power levels are relatively high to compensate for the low integration time used for the simulation (4096 samples per calibrator position, or $34.1\mu\text{s}$ at 120MHz) as opposed to the continuous duty cycle of a real array. A discussion on the trade-off between integration time and calibrator power and what real calibrator power would provide similar performance to the simulated calibrator power accounting for realistic integration time can be found in Section 3.3.

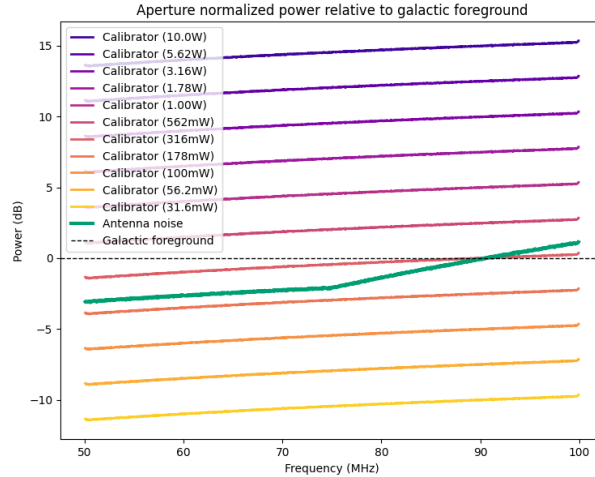


Figure 14: Aperture-normalized average power per antenna, relative to the Galactic foreground, for the calibrator transmitter at each tested power level (10W - 31.6mW), with the calibrator directly above the array at a distance of 100km. The antenna noise contribution used in the simulations is also shown for comparison.

At lower transmitter powers, the quality of ToA estimates degrades, particularly for calibrator positions near the horizon where both increased distance and low antenna gain reduce the received signal strength. To avoid unnecessary computation in regimes where the signal is too weak to yield useable data, a selection criterion is introduced to determine the subset of calibrator positions that provide reliable measurements.

For each candidate calibrator position, ToA estimates are computed for all antennas and compared locally across the array. A ToA measurement is classified as an outlier if it differs by more than two sample bins from all of its neighboring antennas. A calibrator position is considered to provide *good reception* if fewer than 5% of its ToA measurements are classified as outliers. For each transmitter power, the minimum elevation angle above the horizon at which this criterion is satisfied is determined, and only calibrator positions above this angle are included in the reconstruction. The minimum elevation angles, as well as the number of calibrator positions remaining above this elevation angle, for each transmitter power used can be found in Table 1.

Calibrator power	Minimal calibrator elevation angle	Remaining calibrator positions
10.0 W	10°	7619
5.62 W	11°	6897
3.16 W	13°	5725
1.78 W	15°	4767
1.00 W	18°	3683
562 mW	22°	2669
316 mW	26°	1969
178 mW	30°	1500
100 mW	36°	999
56.2 mW	43°	621
31.6 mW	52°	324

Table 1: Calibrator sky coverage for different calibrator power levels.

Using this adaptive selection of calibrator positions, the full reconstruction pipeline is repeated for each transmitter power level. Figure 15 summarizes the impact of transmitter power on position reconstruction performance. Shown are the Root Mean Square Error (RMSE) of the reconstructed antenna positions, as well as for the horizontal and vertical components individually.

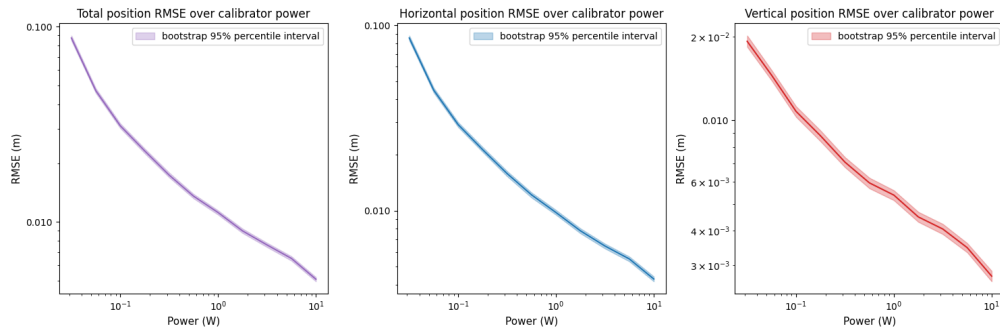


Figure 15: Sensitivity of array reconstruction performance to transmitter power. RMS total (left), horizontal (middle), and vertical (right) position error as a function of transmitter power.

As can be seen, the RMSE initially increases with decreasing calibrator power following a power law. At some point around 100mW, the slope increases. The degradation

in reconstruction accuracy with decreasing transmitter power can be understood as a combination of increased measurement noise and worsening conditioning of the inverse problem. As low SNR calibrator positions are excluded, the effective distribution of calibrator directions becomes more limited, increasing the condition number of the system and amplifying the effect of noise.

Overall, these results show that the proposed method degrades gracefully with decreasing transmitter power and that reliable reconstruction remains possible over a wide range of SNR conditions. The primary limiting factor at low power is not only measurement noise but also the loss of geometric diversity due to the exclusion of low-elevation calibrator positions.

3.3 Integration time trade-off

For a satellite calibrator in lunar orbit, transmitter power is a critical design parameter. Transmission power is constrained by spacecraft limitations such as energy budget and thermal dissipation. It is therefore desirable to operate the calibrator at the lowest possible transmission power while still enabling accurate array calibration.

A simple way to compensate for reduced transmitter power is to increase the integration time used in the time of arrival estimation. Since the received signal is deterministic and known, coherent integration improves the signal-to-noise ratio (SNR) as

$$\text{SNR} \propto P \cdot \sqrt{t_{\text{int}}}, \quad (18)$$

where P is the transmitter power and t_{int} is the integration time.

The simulations in the previous section used an integration time of $t_{\text{int}} = 34.1 \mu\text{s}$. Increasing the integration time to 1 s would yield an SNR improvement factor of

$$\sqrt{\frac{1\text{s}}{34.1 \cdot 10^{-6}\text{s}}} \approx 171, \quad (19)$$

which means that, in theory, the same SNR can be achieved with a transmitter power reduced by a factor of 171.

While this trade-off appears attractive, increasing the integration time has several important drawbacks. First, longer integration times reduce the temporal resolution at which independent calibrator measurements can be obtained. Since the satellite is moving along its orbit, each integration effectively averages over a range of positions. As a result, the effective calibrator sky coverage becomes sparser, reducing the diversity of illumination directions, leading to worse conditioning of the baseline estimation problem.

Second, the usable integration time is coupled to the calibrator motion during integration. For a satellite in low lunar orbit, the apparent angular motion across the sky is non-negligible, and over sufficiently long integration times, this leads to a widening of the correlation peak, reducing timing accuracy. The limitation imposed by calibrator motion can be quantified by considering the time variation of the geometric delay during the integration interval. For antenna i , the propagation delay is

$$\tau_i(t) = \frac{1}{c} \|\vec{x}_i - \vec{x}_s(t)\|. \quad (20)$$

The time derivative of this delay is

$$\dot{\tau}_i = -\frac{1}{c} \hat{n}_i \cdot \vec{v}_s, \quad (21)$$

where $\hat{n}_i$ is the unit vector from the satellite to antenna i , and $\vec{v}_s$ is the satellite's velocity. In far field, $\hat{n}_i \approx \hat{n}$ is approximately constant across antennas, so the relevant quantity is the differential delay rate between antennas:

$$\frac{d}{dt}\Delta\tau_{ij} \approx \frac{1}{cR} (\vec{b}_{ij} \cdot \vec{v}_s), \quad (22)$$

where R is the distance between the array and the satellite, and $\vec{b}_{ij}$ is the baseline vector. This expression shows that delay variations are driven by the ratio v_s/R , the apparent angular velocity of the calibrator.

For a satellite in low lunar orbit, worst-case values, where the satellite is near zenith, are $v_s \sim 1.6\text{km/s}$ and $R \sim 100\text{km}$, giving an angular velocity of

$$\omega \approx \frac{v_s}{R} \sim 1.6 \cdot 10^{-2} \text{ rad/s}. \quad (23)$$

For a representative maximum baseline length of $|\vec{b}_{ij}| \sim 175\text{m}$, the corresponding delay rate is

$$\left| \frac{d}{dt}\Delta\tau_{ij} \right| \sim \frac{|\vec{b}_{ij}|}{c} \omega \approx 9.3 \cdot 10^{-9} \text{s/s} = 9.3\text{ns/s}. \quad (24)$$

To avoid significant degradation of the correlation peak, the change in delay over the integration time should remain small compared to the effective timing resolution. A conservative requirement is that the delay drift remains below a small fraction ϵ of the sampling interval $\Delta t = \frac{1}{f_{\text{sample}}}$. Taking $\epsilon \sim 0.1$ and a sampling interval $\Delta t = \frac{1}{120 \cdot 10^6} \approx 8.3\text{ns}$ yields the constraint

$$t_{\text{int}} \lesssim \frac{\epsilon \Delta t}{d\Delta\tau/dt} \sim \frac{0.83\text{ns}}{9.3\text{ns/s}} \approx 89\text{ms}. \quad (25)$$

This estimate indicates that, for the assumed system parameters, integration times on the order of 0.1 seconds are feasible before motion-induced decorrelation becomes significant. Increasing integration time to 0.1s would yield an SNR improvement of

$$\sqrt{\frac{0.1\text{s}}{34.1 \cdot 10^{-6}\text{s}}} \approx 54. \quad (26)$$

Table 2 shows the equivalent transmitter powers when increasing the integration time from $34.1\mu\text{s}$ to 0.1s.

Original power ($t_{\text{int}} = 34.1\mu\text{s}$)	Equivalent power ($t_{\text{int}} = 0.1\text{s}$)
10.0 W	185 mW
5.62 W	104 mW
3.16 W	58.5 mW
1.78 W	32.9 mW
1.00 W	18.5 mW
562 mW	10.4 mW
316 mW	5.86 mW
178 mW	3.29 mW
100 mW	1.85 mW
56.2 mW	1.04 mW
31.6 mW	0.59 mW

Table 2: Equivalent transmitter power when increasing integration time from $34.1\mu\text{s}$ to 0.1s.

Therefore, while integration times approaching 0.1s are likely achievable without severe degradation, extending significantly beyond this regime provides diminishing returns and may ultimately compromise ToA estimation accuracy. This demonstrates that, while increasing integration time can partially compensate for reduced transmitter power, it cannot be extended arbitrarily. In practice, a balance must be struck between achieving sufficient SNR and maintaining adequate temporal resolution and geometric fidelity of the calibrator measurements. This limits the extent to which transmitter power can be reduced through longer integration alone.

3.4 Effect of static antenna delay

In this subsection, the effect of a static delay offset in one or more of the antennas is investigated. Such delays can arise in practice, for example, due to differences in cable lengths or hardware timing offsets. Unlike geometric delays, these contributions are constant in time and independent of the direction of the incoming signal.

3.4.1 The antenna delay problem

To study this effect, a fixed delay offset is added to all time of arrival (ToA) measurements of antennas, using the extracted ToA dataset from the ideal conditions simulation described in Section 3.1. The full reconstruction pipeline is then applied without modification.

Figure 16 shows the resulting reconstruction errors for a 10ns delay applied to one antenna. A clear systematic deviation is observed, primarily affecting the vertical component of the reconstructed antenna position.

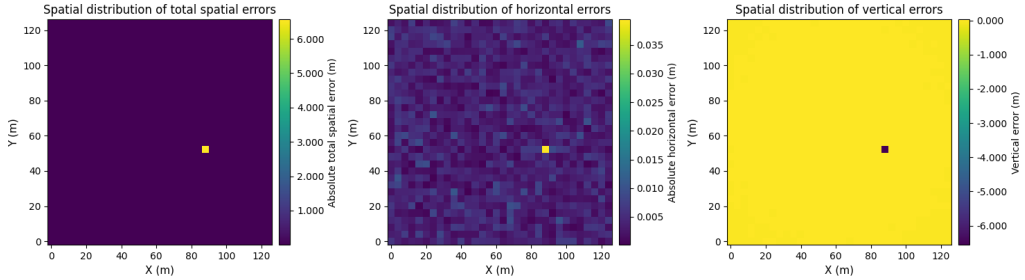


Figure 16: Effect of a static 10ns antenna delay applied to the antenna at (88, 52) on reconstructed array geometry. Spatial maps of absolute total position error (left), absolute horizontal error (center), and vertical error (right).

The origin of this behavior can be understood by examining how a constant delay offset enters the TDoA measurements. If antenna i has an additional delay δ , then all pairwise time differences involving this antenna are shifted by a constant amount:

$$\Delta\tau_{ij} \rightarrow \Delta\tau_{ij} + \delta. \quad (27)$$

In the plane wave approximation, TDoA measurements constrain projections of baseline vectors onto the calibrator direction, $\Delta\tau_{ij} \propto -\vec{s} \cdot \vec{b}_{ij}$. A constant offset therefore acts as an additive bias in these projections, independent of direction. This bias cannot be consistently represented by a true geometric displacement. A physical displacement of an antenna would produce direction-dependent changes in $\Delta\tau_{ij}$, whereas a static delay introduces the same shift for all calibrator directions. As a result, the reconstruction algorithm attempts to absorb the inconsistency by introducing a spurious geometric

offset. Due to the distribution of calibrator directions being approximately uniform horizontally, but vertically only above the array, this manifests primarily as an error in the vertical direction.

3.4.2 Adjusted reconstruction model

Because a static delay is fundamentally inconsistent with the geometric model, it is, in principle, identifiable. To account for this, the reconstruction method can be extended by introducing an additional per-antenna delay parameter δ_i . The measurement model then becomes

$$\tau_i = \frac{1}{c} \|\vec{x}_i - \vec{x}_s\| + \delta_i. \quad (28)$$

The extra delay parameter can be accounted for in the reconstruction method by adding an extra dimension to the baseline vectors and antenna position vectors. Equation 10 then becomes

$$\mathcal{R}(\vec{b}_{ij}) = \sum_k \left[\|\vec{x}_s^{(k)} - \vec{x}_i^{(xyz)}\| - \|\vec{x}_s^{(k)} - (\vec{x}_i^{(xyz)} + \vec{b}_{ij}^{(xyz)})\| + cb_{ij}^t - c\Delta\tau_{ij}^{(k)} \right]^2, \quad (29)$$

where $\vec{b}_{ij} = (\vec{b}_{ij}^{(xyz)}, b_{ij}^t)$ is the baseline vector extended with a differential delay parameter $b_{ij}^t = \delta_i - \delta_j$, and $\vec{x}_i = (\vec{x}_i^{(xyz)}, \delta_i)$. Equation 11 for reconstructing the array geometry remains the same, except now the antenna position vectors $\vec{x}_i$ and baseline vectors $\vec{b}_{ij}$ are in the new extended form.

3.4.3 Reconstruction results

This extended formulation is tested on both the case where a single antenna is given a 10ns delay, and a case where all antennas are assigned a uniformly random delay between 0ns and 10ns. The results can be seen in Figure 17 and Figure 18 respectively.

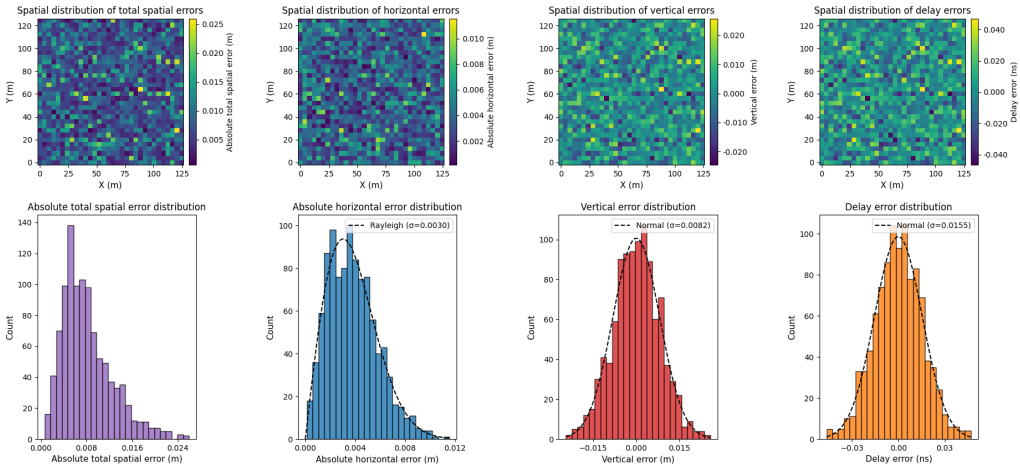


Figure 17: Reconstructed array geometry using the extended model with delay parameters for the case where the antenna at (88, 52) has a fixed 10ns delay. On top: spatial maps of absolute total position error (left), absolute horizontal error (center), and vertical error (right). Below: corresponding error distributions.

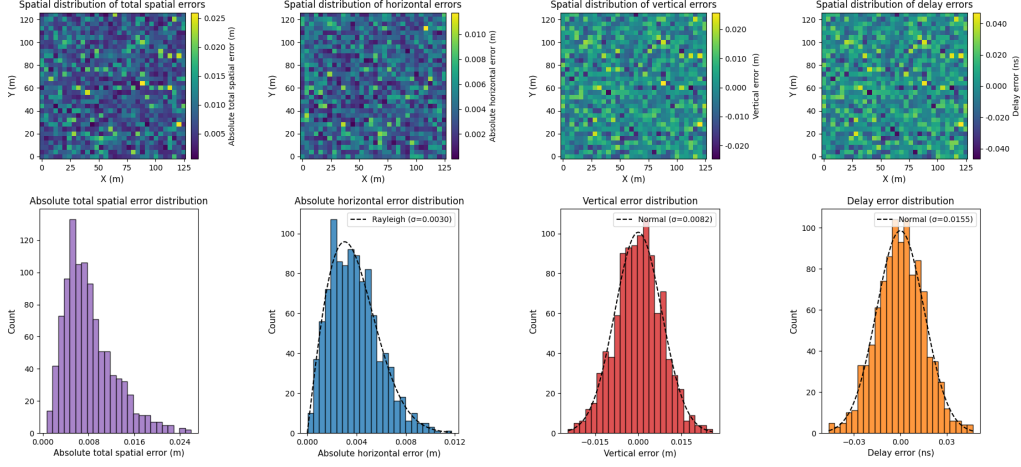


Figure 18: Reconstructed array geometry using the extended model with delay parameters for the case where all antennas are assigned uniformly random delays between 0ns and 10ns. On top: spatial maps of absolute total position error (left), absolute horizontal error (center), and vertical error (right). Below: corresponding error distributions.

The results in Figure 17 and Figure 18 show that the extended reconstruction method successfully removes the systematic bias introduced by static antenna delays. In both cases, the spatial structure observed in Figure 16 is no longer present, and the reconstructed antenna positions are unbiased.

Compared to the ideal case without antenna delays (Section 3.1), an increase in reconstruction noise is observed. In particular, the standard deviation of the vertical error increases by approximately a factor three, while the horizontal components are only weakly affected.

Interestingly, the reconstruction results for the case where a single antenna has a 10ns delay and the case where all antennas have uniformly random delays between 0ns and 10ns are nearly indistinguishable. This indicates that the degradation in reconstruction accuracy is not driven by the magnitude or distribution of the delays themselves. Instead, it is dominated by the presence of the additional delay parameters in the model. The introduction of per-antenna delay parameters increases the dimensionality of the parameter space, introducing a weak degeneracy between geometric and delay parameters. Both affect the measured TDoA as additive contributions, making it difficult to uniquely attribute deviations to either geometric displacement or delays.

The extended formulation is also tested on the case where all antennas are assigned a uniformly random delay between 0ns and 10ns for all measurement sets from Section 3.2. The results can be seen in Figure 19. When compared to the case without delay in Figure 15, the horizontal RMSE is almost unaffected while the effect on the vertical RMSE is increasingly worse with lower transmitter power, going from a factor 3 at 10W to a factor 15 at 31.6mW. While the reduced SNR contributes to an overall loss in precision, it does not by itself explain the increasing discrepancy relative to the case without delay parameters. Instead, the reduced number of measurements limits the amount of data used for the reconstruction, which increases the degeneracy between the geometric parameters and the delay parameters.

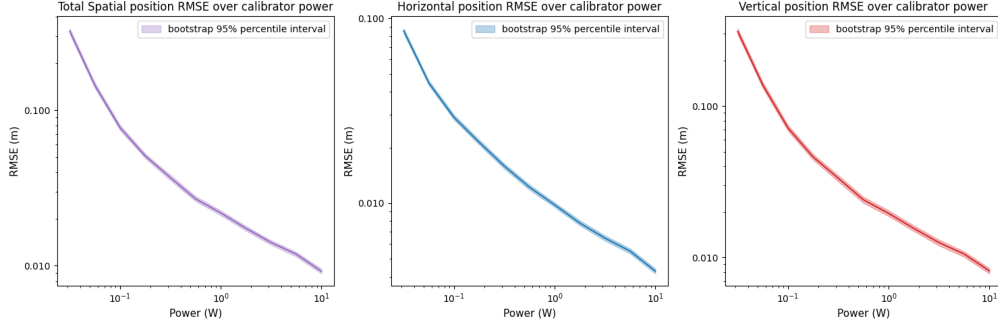


Figure 19: Sensitivity of array reconstruction performance to transmitter power in the presence of uniformly random antenna delays between 0ns and 10ns. RMS total (left), horizontal (middle), and vertical (right) position error as a function of transmitter power.

3.4.4 Summary

These results indicate that static antenna delays can be effectively calibrated within the reconstruction framework by extending the parameter space and that doing so removes systematic bias at the cost of an increase in statistical uncertainty in the vertical component of the position reconstruction. In practice this trade-off is favorable, as unmodeled delays would otherwise lead to significantly larger reconstruction errors.

3.5 Implications for DEX

The reconstruction results from the previous chapter are summarized in Table 3.

$P_{t_{\text{int}}=34.1\mu\text{s}}$	$P_{t_{\text{int}}=0.1\text{s}}$	RMSE _{horizontal}	RMSE _{vertical}	RMSE _{horizontal, delay}	RMSE _{vertical, delay}
10.0 W	185 mW	4.28 mm (1.43 m λ)	2.75 mm (0.92 m λ)	4.28 mm (1.43 m λ)	8.14 mm (2.72 m λ)
5.62 W	104 mW	5.49 mm (1.83 m λ)	3.47 mm (1.16 m λ)	5.50 mm (1.83 m λ)	10.5 mm (3.50 m λ)
3.16 W	58.5 mW	6.46 mm (2.16 m λ)	4.06 mm (1.36 m λ)	6.47 mm (2.16 m λ)	12.5 mm (4.17 m λ)
1.78 W	32.9 mW	7.80 mm (2.60 m λ)	4.49 mm (1.50 m λ)	7.80 mm (2.60 m λ)	15.5 mm (5.18 m λ)
1.00 W	18.5 mW	9.79 mm (3.26 m λ)	5.37 mm (1.79 m λ)	9.78 mm (3.26 m λ)	19.5 mm (6.51 m λ)
562 mW	10.4 mW	12.2 mm (4.06 m λ)	5.94 mm (1.98 m λ)	12.2 mm (4.07 m λ)	24.0 mm (8.02 m λ)
316 mW	5.86 mW	15.9 mm (5.29 m λ)	7.07 mm (2.36 m λ)	15.9 mm (5.29 m λ)	33.3 mm (11.1 m λ)
178 mW	3.29 mW	21.4 mm (7.13 m λ)	8.80 mm (2.93 m λ)	21.4 mm (7.13 m λ)	46.1 mm (15.4 m λ)
100 mW	1.85 mW	29.1 mm (9.71 m λ)	10.8 mm (3.59 m λ)	29.1 mm (9.71 m λ)	71.6 mm (23.9 m λ)
56.2 mW	1.04 mW	44.7 mm (14.9 m λ)	14.5 mm (4.85 m λ)	44.7 mm (14.9 m λ)	137 mm (45.8 m λ)
31.6 mW	0.59 mW	85.3 mm (28.4 m λ)	19.3 mm (6.42 m λ)	85.5 mm (28.5 m λ)	311 mm (103 m λ)

Table 3: From left to right: the calibrator transmission power used in the simulation, the equivalent transmitter power when increasing integration time from 34.1 μ s to 0.1s (3.3), the root mean square error in array element location reconstruction in the horizontal and vertical directions (3.2), the root mean square error in array element location reconstruction in the horizontal and vertical directions, including per-element delay parameters and using the extended reconstruction method (3.4). RMSE values are expressed both in millimeters and in milli λ , where $\lambda = 2.998\text{m}$ is the wavelength of 100MHz radiation.

The results in Table 3 can be directly evaluated against the tolerance criteria established for DEX [25]. In this work, position uncertainties are expressed as a fraction of the observing wavelength, with a critical threshold of 0.027λ (0.081m at $f = 100\text{MHz}$) required to limit visibility errors and suppress significant foreground leakage into the 21-cm window.

For the case without delay terms, both horizontal and vertical reconstruction errors remain below this threshold across all but the weakest considered transmitter power, with fractional uncertainties at the level of a few $\text{milli}\lambda$. In this regime, phase errors are expected to remain sufficiently small such that the corresponding impact on the power spectrum is acceptable for 21-cm cosmology.

In contrast, when per antenna delays are included in the reconstruction, the horizontal errors remain within the acceptable tolerance regime, but the vertical errors exceed the 0.027λ threshold for calibrator powers below 100mW (1.85mW compensated for realistic integration time). Position errors at this level would lead to increased foreground leakage beyond the levels considered acceptable for precision 21-cm measurements. This would have to be corrected for with the inclusion of more calibrator positions.

4 Discussion

The results presented in this work demonstrate that antenna array geometry can be reconstructed with high accuracy using TDoA measurements from an orbital calibrator. However, several simplifying assumptions were made in the modeling and simulation framework. Evaluating the impact of these assumptions is essential for understanding how the method will perform under real-world conditions.

4.1 Beam pattern assumption

A constraint in the simulation is that all antennas share identical, perfectly known beam patterns. In practice, individual antennas will exhibit beam pattern variations due to deployment and surface irregularities. Directional gain and phase errors directly affect not only geometry reconstruction performance but also introduce artifacts in the measured signals, which can mimic and obscure the 21-cm signal.

A critical question is whether beam pattern errors can be disentangled from geometric errors. Since beam effects primarily affect gain while geometric errors primarily affect phase, they may be largely orthogonal. However, in low SNR regimes or with incomplete sky coverage, strong degeneracies may arise. Extending the reconstruction framework to jointly estimate beam patterns and geometry is a valuable next step. The simulation architecture developed in this work is well suited for this, as it already performs frequency domain synthesis, and this step can be easily extended to include beam variations.

4.2 Satellite Doppler effects

The current simulation treats the calibrator as stationary during a sample window, neglecting Doppler shifts. In reality, the relative velocity between the satellite and the array introduces a frequency shift, which can reduce correlation peak sharpness and degrade ToA estimation accuracy.

Because the transmitted waveform is known, this effect is, in principle, correctable. A Doppler-compensated correlation could be implemented by searching over a Doppler parameter space. Incorporating this into the reconstruction pipeline would allow longer integration times and improved SNR without sacrificing timing precision.

4.3 E-field interpolation onto a regular grid

The reconstruction method assumes that, once antenna positions are known, the measured electric field samples can be interpolated back onto an ideal grid with sufficient accuracy to enable FFTT imaging. This step is not explicitly modeled in the current work but is critical for FFTT operation.

The accuracy requirements for interpolation depend on the relationship between positional errors and phase errors. If interpolation errors introduce phase distortions comparable to or larger than those from residual geometry errors, they become the bottleneck. In this case, further improvements in geometry reconstruction would not translate into improved imaging performance. A combined analysis of geometry reconstruction and interpolation error propagation would provide a more complete assessment of system performance.

4.4 Other model assumptions

Several additional assumptions may impact real-world performance:

- Perfect knowledge of calibrator position and timing;
- Perfectly stable and synchronized clocks for sampling;
- Absence of polarization effects;
- Absence of multipath effects.

In practice, uncertainties in the satellite’s position, clock stability, synchronization, and environmental factors can introduce additional systematic effects that are not included in the present simulation framework. These effects may bias the inferred delays used during calibration, thereby degrading the accuracy of the reconstructed antenna geometry and the resulting imaging performance.

Errors in the assumed calibrator position affect the predicted geometric delays across the array. Since the reconstruction relies on matching measured and expected arrival times, such uncertainties will lead to uncertainties in the reconstructed antenna positions, introducing geometric distortions. It may be possible to remove the dependence on perfect knowledge of the calibrator position from the reconstruction, as information of the calibrator position is embedded in the measured ToA. Whether this is possible in practice needs to be investigated.

Clock instability and synchronization errors similarly introduce additional timing offsets. Slow clock drifts could mimic geometric delay structure, while short-timescale jitter broadens correlation peaks, reducing ToA precision. In large arrays, differential clock errors may therefore reduce the effectiveness of the calibration approach.

The simulation pipeline currently ignores polarization effects. The beam pattern currently assumes a symmetric loss of sensitivity towards the local horizon in all directions. For real polarization sensitive antennas, this fall-off can be anisotropic. Sensitivity degradation occurs primarily along the antenna’s polarization axis, while response in the perpendicular direction can remain significantly stronger. Consequently, incorporating polarization effects may actually improve SNR for low-elevation calibrator positions in certain directions.

Multipath propagation introduces additional reflected signals that interfere with the direct waveform. These effects are direction- and frequency-dependent and can modify the effective antenna response and phase structure. In practice, this behavior may be absorbed into beam pattern calibration terms.

5 Conclusion

This thesis presented a method for reconstructing the geometry of antenna arrays deployed for Fast Fourier Transform Telescopes (FFTT's) on the lunar far side. The approach uses Time Difference of Arrival measurements from a known calibrator signal transmitted by a satellite in low lunar orbit, enabling the inference of antenna positions without direct physical measurement.

A complete simulation framework was developed to evaluate the method under realistic observing conditions, including sky noise, instrumental effects and calibrator signal. The reconstruction process combines a computationally efficient plane wave approximation with a realistic spherical wavefront model for high accuracy.

Under ideal conditions, the method achieves millimeter scale accuracy in antenna position reconstruction. The analysis shows that:

- The plane wave approximation introduces systematic bias, which is effectively removed with the spherical wave model.
- Reconstruction performance degrades gradually with decreasing transmitter power, with decreased SNR, fewer included data points, and reduced calibrator sky coverage contributing to increased errors.
- Increasing integration time can partially compensate for reduced transmitter power but is limited by satellite motion and, in extreme cases, loss of temporal resolution.
- Static antenna delays can be incorporated into the model, eliminating bias but increasing position reconstruction uncertainty in the vertical direction.

When evaluated against the positional accuracy requirements for the DEX mission, the method satisfies tolerance criteria over a wide range of operating conditions. This supports the feasibility of using an orbital calibrator for in-situ array calibration in lunar FFTT systems.

The results highlight that accurate geometry reconstruction is achievable without additional surface infrastructure, making it well-suited for large, autonomously deployed arrays. Future work should focus on extending the model to include beam pattern uncertainties and Doppler effects, and investigate the accuracy of E-field interpolation.

Overall, it is demonstrated that array geometry reconstruction via an orbital transmitter is a viable solution for next-generation experiments targeting the cosmological 21-cm signal.

References

- [1] Jack O. Burns. Transformative science from the lunar farside: observations of the dark ages and exoplanetary systems at low radio frequencies. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 379(2196), 2021.
- [2] Steven Furlanetto, S. Peng Oh, and Frank Briggs. Cosmology at low frequencies: The 21 cm transition and the high-redshift universe. *arXiv preprint*, 2006.
- [3] Adrian Liu and J. Richard Shaw. Data analysis for precision 21 cm cosmology. *Publications of the Astronomical Society of the Pacific*, 132, 2020.
- [4] Jonathan R Pritchard and Abraham Loeb. 21 cm cosmology in the 21st century. *Reports on Progress in Physics*, 75(8):086901, jul 2012.
- [5] N. Barry, B. Hazelton, I. Sullivan, and M. F. Morales. Calibration requirements for detecting the 21 cm epoch of reionization power spectrum. *Monthly Notices of the Royal Astronomical Society*, 461:3135–3144, 2016.
- [6] A. Datta, C. M. Trott, and R. B. Wayth. The impact of ionospheric refraction on epoch of reionization power spectra. *Astrophysical Journal*, 833:119, 2016.
- [7] A. E. E. Rogers and J. D. Bowman. Absolute calibration of a wideband antenna and spectrometer for measuring the global 21 cm signal. *Radio Science*, 50:130–146, 2015.
- [8] S. Singh, R. Subrahmanyam, N. Udaya Shankar, and A. Raghunathan. Saras 2: A spectral radiometer for probing cosmic dawn and the epoch of reionization. *Astrophysical Journal*, 858:54, 2018.
- [9] L. Philip et al. Probing radio intensity at high-z from marion (prizm): Design and commissioning. *Journal of Astronomical Instrumentation*, 8:1950004, 2019.
- [10] D. C. Price, L. J. Greenhill, A. Fialkov, et al. Design and characterization of the large-aperture experiment to detect the dark age (leda) radiometer systems. *Monthly Notices of the Royal Astronomical Society*, 478:4193–4213, 2018.
- [11] J. D. Bowman, A. E. E. Rogers, R. A. Monsalve, T. J. Mozdzen, and N. Mahesh. An absorption profile centred at 78 megahertz in the sky-averaged spectrum. *Nature*, 555:67–70, 2018.
- [12] Junsong Cang, Andrei Mesinger, Steven G. Murray, Daniela Breitman, Yuxiang Qin, and Roberto Trotta. The edges measurement disfavors an excess radio background during the cosmic dawn. *Astronomy and Astrophysics*, 698:A152, June 2025.
- [13] M. P. van Haarlem et al. Lofar: The low-frequency array. *Astronomy & Astrophysics*, 556:A2, 2013.
- [14] D. R. DeBoer, A. R. Parsons, J. E. Aguirre, et al. Hydrogen epoch of reionization array (hera). *Publications of the Astronomical Society of the Pacific*, 129:045001, 2017.
- [15] P. E. Dewdney, W. Turner, R. Millenaar, R. McCool, T. Lazio, and T. Cornwell. Ska1 system baseline design. *SKA Organisation Technical Report*, 2013. SKA-TEL-SKO-DD-001.

- [16] J. R. Pritchard et al. Cosmology from the epoch of reionization and cosmic dawn with the ska. *Proceedings of Science (AASKA14)*, page 012, 2015.
- [17] A. Datta, J. D. Bowman, and C. L. Carilli. Foreground subtraction and the wedge in redshifted 21 cm power spectra. *Astrophysical Journal*, 755:90, 2012.
- [18] M.J. Bentum, C.J.M. Verhoeven, and Albert Boonstra. Olfar, orbiting low frequency antennas for radio astronomy. 01 2009.
- [19] Jack O. Burns, Gregg Hallinan, Jim Lux, Lawrence Teitelbaum, Jonathon Kocz, Robert MacDowall, Richard Bradley, David Rapetti, Wenbo Wu, Steven Furlanetto, Alex Austin, Andres Romero-Wolf, Tzu-Ching Chang, Judd Bowman, Justin Kasper, Marin Anderson, Zhongwen Zhen, Jonathan Pober, and Jordan Mirocha. Nasa probe study report: Farside array for radio science investigations of the dark ages and exoplanets (farside). *arXiv preprint*, 2019.
- [20] Dayton L. Jones, T. Joseph W. Lazio, and Jack O. Burns. Dark ages radio explorer mission: Probing the cosmic dawn. *arXiv preprint*, 2014.
- [21] C.D. Brinkerink, M.J. Arts, M.J. Bentum, A.J. Boonstra, B. Cecconi, A. Fialkov, J. García Gutiérrez, S. Ghosh, J. Grenouilleau, L.I. Gurvits, M. Klein-Wolt, L.V.E. Koopmans, J. Lazendic-Galloway, Z. Paragi, D. Prinsloo, R.T. Rajan, E. Rouillé, M. Ruiter, J.A. Tauber, H.K. Vedantham, A. Vecchio, C.J.C. Vertegaal, J.C.F. Zandboer, and P. Zucca. The dark ages explorer (dex): a filled-aperture ultra-long wavelength radio interferometer on the lunar far side. *arXiv preprint*, 2025.
- [22] Max Tegmark and Matias Zaldarriaga. The fast fourier transform telescope. *arXiv preprint*, 2009. Submitted 29 May 2008 (v1); revised 24 April 2009 (v2).
- [23] Emi Tamura, Jack Fried, Sven Herrmann, Paul O’Connor, Eric J. Raguzin, and Anže Slosar. Design and characterization of the engineering model of the spectrometer onboard lusee-night. *Radio Science*, 59, 2024. LuSEE-Night: the lunar surface electromagnetics experiment for low-frequency radio astronomy.
- [24] Nithyanandan Thyagarajan, Adam P. Beardsley, Judd D. Bowman, and Miguel F. Morales. A generic and efficient e-field parallel imaging correlator for next-generation radio telescopes. *Monthly Notices of the Royal Astronomical Society*, page stx113, January 2017.
- [25] S Ghosh, L V E Koopmans, C Brinkerink, A R Offringa, A J Boonstra, S A Brackenhoff, E Ceccotti, J K Chege, L Y Gao, B K Gehlot, L I Gurvits, C Höfer, F G Mertens, M Mevius, S Munshi, A Saxena, J A Tauber, H Vedantham, S Yatawatta, and S Zaroubi. The impact of lunar topography on the 21-cm power spectrum for grid-based arrays: Insights for the dark-ages explorer (dex). *Monthly Notices of the Royal Astronomical Society*, 546(3), January 2026.
- [26] Jordi C.F. Zandboer. Active antenna design for lunar detection of global 21 cm-signals from the dark ages. Master’s thesis, Eindhoven University of Technology, September 13 2023. Supervisor: David Prinsloo.
- [27] Danny Price. Python interface for global diffuse sky models. <https://github.com/telegraphic/pygdsmtree/master>, 2024. Accessed: 2025-06-09.
- [28] Haoxuan Zheng, Max Tegmark, Joshua S. Dillon, Doyeon A. Kim, Adrian Liu, Abraham Neben, Justin Jonas, Patricia Reich, and Wolfgang Reich. An improved model of diffuse galactic radio emission from 10 mhz to 5 thz. *arXiv preprint*, 2016.

- [29] Sarod Yatawatta. Sagecal radio interferometry simulation and calibration software. <https://github.com/nlesc-dirac/sagecal>, 2024. Accessed: 2025-05-21.
- [30] Sam van den Ende. Fringe radio interferometry per-antenna simulation software. <https://github.com/samvandenende/fringe>, 2026. Accessed: 2026-06-28.
- [31] Sarod Yatawatta. Sagecal patch for use of lunar reference frame for alo. <https://github.com/nlesc-dirac/sagecal/pull/206>, 2024. Accessed: 2025-05-21.
- [32] A. Richard Thompson, James M. Moran, and George W. Swenson. *Interferometry and Synthesis in Radio Astronomy*. Springer, Cham, Switzerland, 3rd edition, 2017.

A Assessment of SAGECAL for Fast Fourier Transform Telescope Signal Simulation

The use of an industry-standard radio-interferometric array simulation tool, namely **SAGECAL**[29] was investigated. **SAGECAL** can use complex user-defined sky models and custom array configurations. It even includes an ALO-specific patch[31] to support lunar reference frames.

The problem with using **SAGECAL** for this research is with the data products it produces. In a conventional radio interferometer, the fundamental observable is the visibility, which can be defined as the time-averaged complex correlation between the electric fields measured by a pair of antennas. For antennas i and j , the visibility at frequency f is given by

$$V_{ij}(f) = \left\langle E_i(f) E_j^*(f) \right\rangle_t, \quad (30)$$

where:

- $E_i(f)$ is the complex electric field measured at antenna i ;
- $E_j^*(f)$ is the complex conjugate of the electric field measured at antenna j ;
- $\left\langle \cdot \right\rangle_t$ is a time averaging operator.

The measured visibilities relate directly to the Fourier transform of the sky brightness distribution through the van Cittert–Zernike theorem [32]. Fast Fourier Transform Telescopes (FFTTs) differ in this aspect. They do not explicitly compute visibilities. Instead, they operate directly on per-antenna electric field measurements using spatial FFTs.

Industry-standard simulation tools, including **SAGECAL**, operate fundamentally on visibilities, simulating them directly rather than simulating per-antenna measurements and calculating visibilities from there. This saves on computation time, but this approach is incompatible with the data processing pipeline of FFTTs. Furthermore, in order to extract time of arrival (ToA) information from the calibrator signal, the raw per-antenna time domain E-field samples are required. Due to the time averaging found in the calculation of visibilities, information about instantaneous per-antenna E-field measurements is lost.

B Simulation verification

A variety of validation tests of the simulation tool were performed. All tests are performed using a regular 32×32 square antenna array with 4m inter-element spacing. The antennas are configured to have a 120MHz sampling frequency, a mixing frequency of 45MHz, an observing bandwidth from 5MHz to 55MHz, and sampling windows containing 4096 samples. The goal of these tests is to verify that the simulation produces physically consistent and scientifically accurate outputs when compared against a trusted reference simulator (**SAGECAL**), as well as to confirm correct behavior in Fast Fourier Transform imaging.

B.1 Imaging comparison with SAGECAL

The visibilities are turned into images using **WSCLEAN** with size 256 256, scale 0.25deg, niter 1000, mgain 0.8 and auto-threshold 3. Note that this test focuses on comparing results between the custom simulation and **SAGECAL**, not on the quality of the images.

Single source in zenith

In this test, a single point source is placed in the zenith of the array with a spectral flux density of 10Jy. The images produced by WSCLEAN can be seen in Figure 20. In this simple test case, the resulting images are virtually identical, both showing a single source in the center with a spectral flux density of 10Jy.

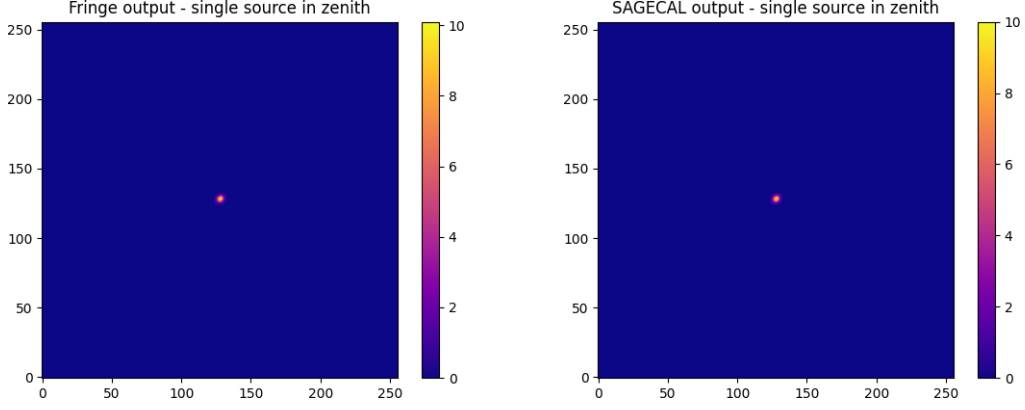


Figure 20: Comparison of images produced from simulated visibilities for a single 10Jy point source located at zenith. Left: custom simulation. Right: SAGECAL reference simulation.

Single source off-center

In this test, a single 10Jy point source is placed with direction vector $\|(1,1,1)\| = \frac{1}{\sqrt{3}}(1,1,1)$. The images produced by WSCLEAN can be seen in Figure 21. In this case, both simulations show a source in the same location, but where SAGECAL shows a 10Jy source, the custom simulation shows a $\frac{10}{3}$ Jy source. This is exactly what is expected, as the custom simulation includes a $\cos^2(\theta)$ antenna gain pattern that works out to a factor $(\frac{1}{\sqrt{3}})^2 = \frac{1}{3}$ and this gain pattern is absent in the SAGECAL simulation.

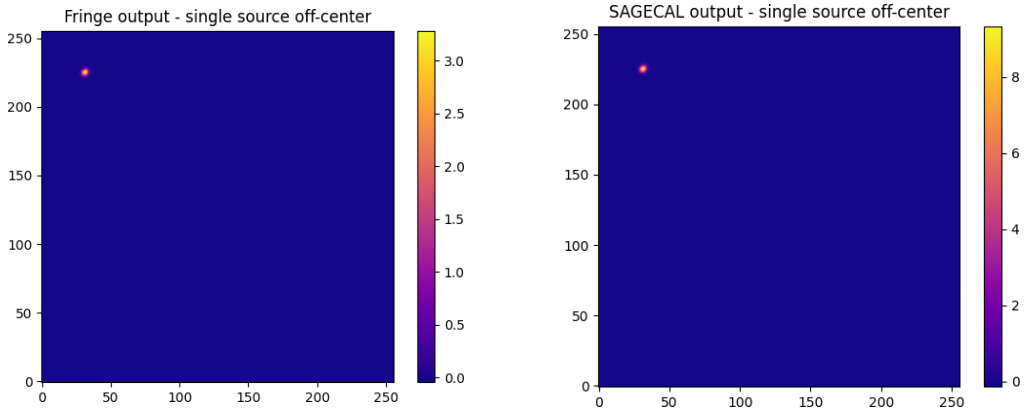


Figure 21: Comparison of images for a single 10Jy point source located off-zenith in direction $\frac{1}{\sqrt{3}}(1,1,1)$. Left: custom simulation. Right: SAGECAL reference simulation.

Two sources

In this test, the two sources from the previous two tests are combined in one simulation. The images produced by WSCLEAN can be seen in Figure 22. This result shows that the custom simulation can accurately handle simulations with multiple sources.

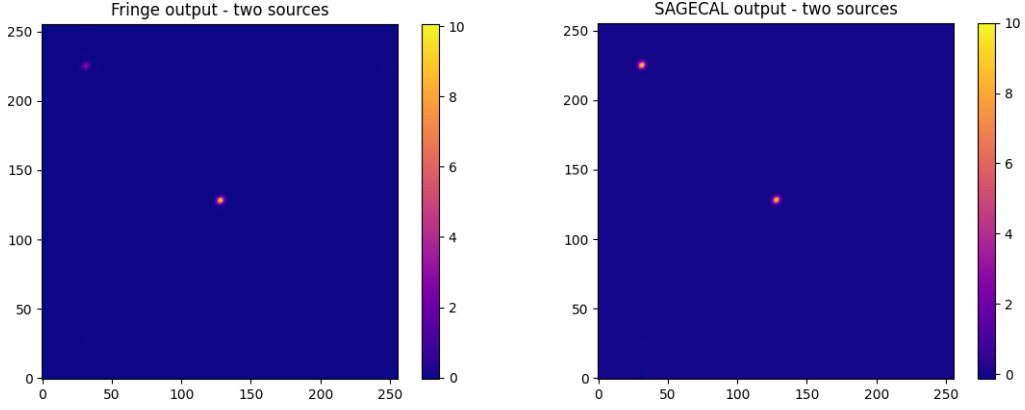


Figure 22: Comparison of images for a two-source sky model combining a 10Jy zenith source and a 10Jy off-zenith source in direction $\frac{1}{\sqrt{3}}(1, 1, 1)$. Left: custom simulation. Right: SAGECAL reference simulation.

Realistic sky

In this test, an accurate sky model was generated as explained in Section 2.3.2 and used for simulation. To facilitate direct comparison between the output images from both simulations, the spectral flux density of the sky sources used by the custom simulation has been increased by a factor $\cos^2(\theta)$, negating the antenna gain pattern difference. The images produced by WSCLEAN can be seen in Figure 23. The output images contain aliasing and imaging artifacts, but importantly, the images from the custom simulation data and from SAGECAL data appear virtually identical. This indicates that, even for complex sky models, the data produced by the custom simulation is qualitatively on par with SAGECAL.

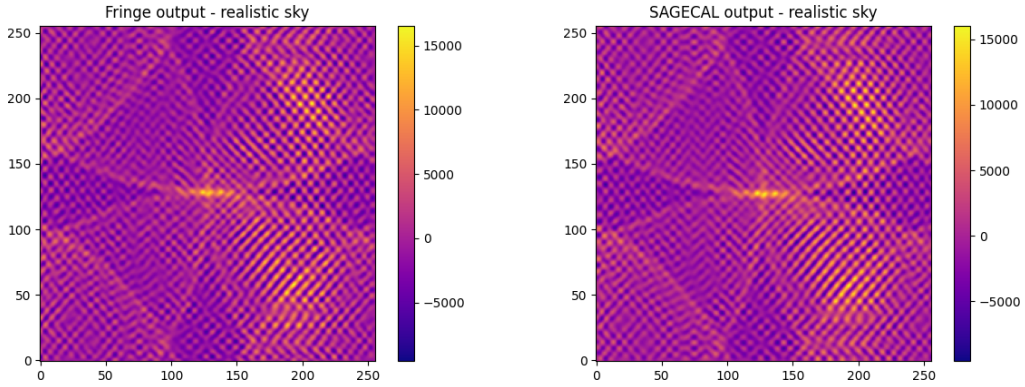


Figure 23: Comparison of dirty images for a realistic sky model. Left: custom simulation. Right: SAGECAL reference simulation. The input sky model for the custom simulation has been scaled by $\cos^2(\theta)$ to compensate for the antenna gain pattern.